

OBJECTIVES

By studying this chapter, the reader should be able to:

- Identify the major components of a nuclear counting system and describe their properties and functions.
- Distinguish between differential and integral sorting in a pulse height analyzer.
- Define the advantages of a multichannel analyzer over a single-channel analyzer.
- Identify the factors contributing to determinate errors in radioactivity measurements.
- Discuss the property of resolving time of radiation detectors, and differentiate between nonparalyzable and paralyzable systems.
- Explain how a pulse-height spectrum is acquired, and identify the mechanisms that give rise to various peaks on a spectrum.
- Delineate the properties of a photopeak on a pulse-height spectrum, and explain the mechanism of photopeak counting.
- Describe the various properties of a radioactive pharmaceutical that are important to applications in nuclear imaging.

INTRODUCTION

Signals from a radiation detector used to detect and measure radioactivity are usually voltage pulses formed by collecting electrical charge or light flashes in the detector. Ideally, each voltage pulse corresponds to the emission of a single ionizing particle or γ -ray photon from the radioactive sample. The pulses may be transmitted from the detector to a succession of electronic circuits for analysis and display. With some detectors, the amplitude of the pulses depicts the energy deposited in the detector during interactions of the radiation. The pulses can be selectively registered if a discriminator or pulse height analyzer is present in the electronic circuitry. Data displayed by output devices must be interpreted in terms of their statistical significance and their relationship to the activity and modes of decay of the radioactive sample. Some of the difficulties encountered during this interpretation are considered in this chapter.

COUNTING SYSTEMS

Counting systems are assembled by combining various electronic circuits and display devices. A general-purpose counting system is outlined in Figure 9-1. Each component of this system is discussed separately in the following sections.

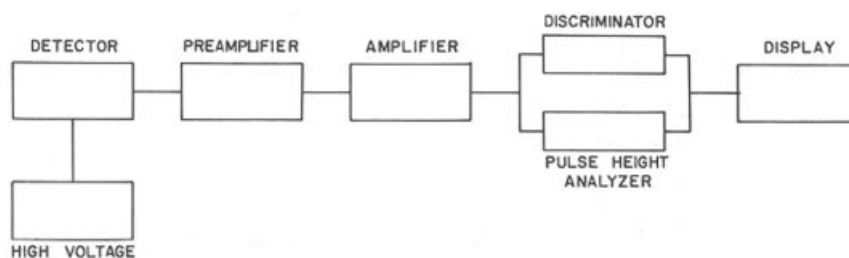


FIGURE 9-1
Components of a general-purpose counting system.

Preamplifiers

Almost all radiation detectors exhibit low capacitance and high impedance. Signals from these detectors are distorted and severely attenuated if they are transmitted by coaxial cable directly from the detector to an amplifier some distance away. To reduce these effects, a preamplifier may be inserted near the detector. The purpose of the preamplifier is to match the impedance of the detector with that of the amplifier. When this is done, the cable to the amplifier may be several feet long. Preamplifiers are used also to “clip” and “shape” the voltage pulse from the detector to meet the specifications of the amplifier.

The term *preamplifier* is misleading when applied to impedance matching circuits, because the voltage pulse from the detector usually is not amplified by the circuit. In fact, the gain of some preamplifiers is less than unity. Other preamplifiers may have a gain as high as 10^4 or 10^5 .

Amplifiers

An amplifier is used to increase the size and vary the shape of signals from the detector and preamplifier. An amplifier may be either voltage sensitive or charge sensitive, depending on the type of signal received at the input terminal. The increase in signal size is described as the *amplifier gain*, which is the ratio of the height of the voltage pulse leaving the amplifier to the size of the signal received at the input terminal of the amplifier. Depending on the type of detector and the characteristics of circuits in the counting system, an amplifier gain of 1000 to 10,000 may be required.

The pulse furnished by a typical amplifier is shown in the margin. The *pulse rise time* is the time required for the pulse amplitude to increase from 10% to 90% of its maximum amplitude. The *pulse decay time* is the time required for the pulse to decrease from maximum amplitude to 10% of maximum. The rise time of an amplifier should be less than the time required to collect the ion pairs or light produced during interaction of a particle or photon in the detector. The voltage pulse should be terminated rapidly to prevent the amplifier from summing successive pulses from the detector. The *integration time* of an amplifier is the time required to form an output pulse. The integration time is a compromise between the time required for complete collection of a signal from the detector and the time that causes a significant number of successive pulses to combine. The integration time is significantly less than $1\ \mu\text{s}$ for amplifiers used with NaI(Tl) scintillation detectors.

Large input pulses or high rates of reception of input pulses may cause the characteristics of an amplifier to change. For example, a few output pulses may be distorted after a very large pulse (e.g., a pulse produced by interaction of a cosmic ray) has been received by the amplifier. This distortion of output pulses is termed *pulse-amplitude overloading*. *Count-rate overloading* refers to pulse-shape distortion caused by delivery of pulses to the amplifier at too high an input rate. Pulse-amplitude overloading and count-rate overloading may severely distort the data displayed by the output device.¹

Pulses are amplified linearly in most amplifiers used in counting circuits. Linear amplification may be a handicap if pulses from the detector are widely variable in size. For these applications, an amplifier with logarithmic gain may be useful. With a logarithmic amplifier, the size of the output pulse is proportional to the logarithm of the size of the input pulse. In such an amplifier, a wide range of input pulses may be amplified without pulse-amplitude overloading and without rejection of very small pulses.

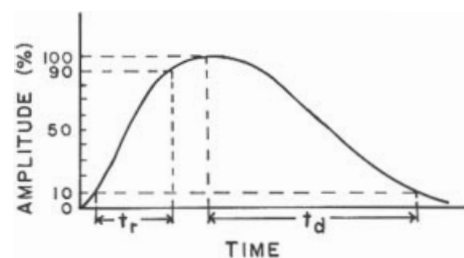
Pulse Height Analyzers

With a detector such as a scintillation or semiconductor detector, the height of a voltage pulse from the detector (and amplifier) is proportional to the energy expended in the detector by a charged particle or an x or γ ray. A typical train of pulses from

A high impedance means that a detector has a high internal resistance to the flow of electrons.

Impedance matching means that the preamplifier presents a high impedance input to the detector and a low impedance output to succeeding electronic components. With this arrangement, signals from the detector are transmitted to the counting electronics in an undistorted fashion.

$$\text{Gain} = \frac{\text{Size of output signal}}{\text{Size of input signal}}$$



MARGIN FIGURE 9-1

Pulse delivered by an amplifier. Amplitude may be either voltage or instantaneous current. Pulse rise time t_r and pulse decay time t_d are shown.

Performance criteria of an amplifier in a nuclear counting circuit include:

- Signal amplification
- Accurate measurement of the detector signal
- Acceptable performance at high count rates
- Maximum signal-to-noise ratio
- Rejection of pulse pile-up at high count rates

These criteria are not independent, and choices among them are required for any particular counting system.

A single-channel pulse-height analyzer is also known as a differential discriminator.



MARGIN FIGURE 9-2

Train of voltage pulses emerging from an amplifier that follows a preamplifier and a scintillation or semiconductor detector. The height of each pulse reflects the energy expended in the detector during interaction of an incident photon or particle.

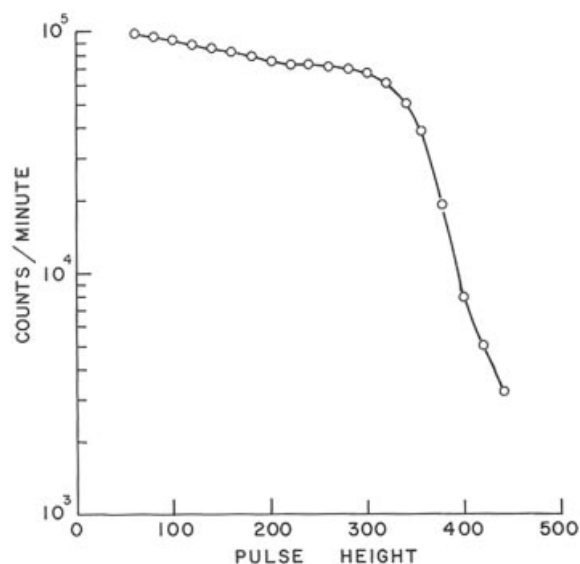


FIGURE 9-2

Integral pulse height spectrum for ^{131}I measured with a 2×2 -in. NaI(Tl) well crystal.

an amplifier connected to a scintillation or semiconductor detector is depicted in the margin. These pulses may be sorted by a single-channel pulse height analyzer (PHA) to yield a pulse height spectrum. This spectrum reflects the distribution of energies deposited in the detector during interactions of incident photons or particles. Two techniques, *differential sorting* and *integral sorting*, are used for pulse height analysis.

For integral sorting, a single discriminator in the pulse height analyzer is varied from a position where all pulses are transmitted to the display device, to a position where all pulses are rejected. Shown in Figure 9-2 is an integral spectrum for ^{131}I . At any value of pulse height on the x axis, the height of the curve denotes the number of pulses that are large enough to pass the input discriminator and reach the display device.

A single-channel pulse height analyzer is composed basically of two discriminators connected to an anticoincidence circuit. The discriminators transmit pulses above a certain minimum size. In Figure 9-3, for example, the lower discriminator transmits pulses larger than size V_1 , and the upper discriminator transmits pulses larger than size V_2 . Pulses from the amplifier are applied simultaneously to both discriminators. Pulses too small ($< V_1$) to be transmitted by either discriminator are rejected and not transmitted to the anticoincidence circuit. Pulses of size between V_1 and V_2 are transmitted by the lower discriminator only and are delivered to one input terminal of the anticoincidence circuit. Pulses large enough ($> V_2$) to be transmitted by both discriminators are delivered simultaneously to both input terminals of the anticoincidence circuit. The anticoincidence circuit transmits a pulse to the display device when it receives a pulse at only one input terminal. A pulse is not transmitted to the display device when signals are received simultaneously at both input terminals.

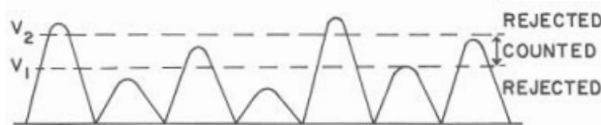


FIGURE 9-3

Diagram of a series of voltage pulses impinging on discriminators V_1 and V_2 of a single-channel pulse height analyzer. Pulses with height between V_1 and V_2 are counted. Pulses with height less than V_1 are rejected by both discriminators, and pulses with height greater than V_2 are rejected by an anticoincidence circuit.

Consequently, the display device registers only the number of pulses of size between V_1 and V_2 . The range of pulse sizes registered by the display device may be varied by changing the settings V_1 and V_2 of the lower and upper discriminators. These settings may be labeled on the pulse height analyzer as “lower discriminator” and “upper discriminator,” “ E_1 ” and “ E_2 ,” or “lower level” and “upper level.” Occasionally (e.g., in some scintillation cameras), the lower and upper discriminators of a pulse height analyzer are not adjustable. With this type of analyzer, differential pulse height analysis may be achieved by changing the range of pulse sizes emerging from the detector or by varying the amplification of the pulses in the amplifier. These changes may be accomplished by varying the high voltage of the photomultiplier tube in a scintillation detector, or by changing the gain of the amplifier.

In a pulse height analyzer with discriminators that can be adjusted independently of each other, the range of pulse sizes transmitted to the anticoincidence circuit may be affected severely by small fluctuations in voltage applied to the discriminators. To reduce this dependence on voltage stability, the upper discriminator may be arranged to “ride” on the lower discriminator. In this manner, a constant difference in pulse size may be maintained between the discriminators. The lower discriminator may be termed “window level,” “threshold,” “ E ,” “baseline,” or “peak voltage,” and the difference in pulse size between the two discriminators may be referred to as “window width,” “window,” “slit width,” “ ΔE ,” or “percent window.” The position of the lower discriminator determines the minimum size of pulses transmitted to the display device, and the width of the window determines the increment of pulse sizes transmitted. As the window width is reduced, fewer pulses are transmitted to the display device, but the pulse-size (energy) resolution of the pulse height spectrum is increased. A differential pulse height spectrum for ^{131}I is shown in Figure 9-4.

Most pulse height analyzers may be operated in either the integral or differential mode. In the differential mode, some analyzers may be operated with independent lower and upper discriminators or with a variable lower discriminator and a dependent window.

The *linearity* of a pulse height analyzer describes the relationship between the position of the lower discriminator (or middle of the window) and the size of pulses transmitted to the display device. For a counting system with linear amplification, a straight line should be obtained if the average size of pulses admitted to the display

The range of pulse sizes displayed on the output device may be varied by changing the following:

- Detector high voltage
- Amplifier gain
- Lower and upper discriminators

In 1927, Niels Bohr described the Principle of Complementarity for quantum systems. As a companion to Heisenberg’s Uncertainty Principle, the Principle of Complementarity states that for every measurable quantity there exists a “complementary” measurable quantity. Increasing the accuracy of one quantity reduces the accuracy with which the complementary quantity can be known.

At the end of the lecture in which Bohr presented the Principle of Complementarity, a member of the audience asked “What, then, is complementary to Truth?” Without hesitation, Bohr responded “Clarity!”

Reported by L. Litt, M.D.
New York Times, May 2000

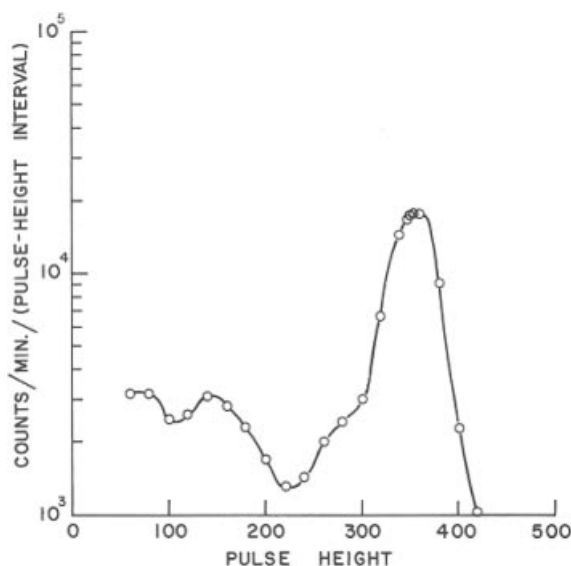


FIGURE 9-4

Differential pulse height spectrum for ^{131}I measured with a 2×2 -in. NaI(Tl) well crystal. The peak in the spectrum represents pulses produced by total absorption of 364-keV γ rays in the scintillation crystal.

Pulses selected by a pulse-height analyzer for measurement or image display are referred to as “Z pulses.”

The multichannel analyzer began to replace single-channel analyzers in most counting systems in the late 1960s.

In a scintillation camera, pulses transmitted by the pulse-height analyzer may be counted in a scaler, displayed on a cathode-ray tube, recorded on film, transmitted to a computer, stored on magnetic tape or laser disk, or any combination thereof.

Determinate (also termed systematic) errors are caused by inadequate design of the measurement system. Determinate errors are distinguishable from indeterminate (random) errors that are intrinsic to measurements of radioactivity.

Detectors such as whole-body counters used to measure small amounts of radioactivity must be well shielded to reduce the background count rate to very low levels. Pre-World War II steel from decommissioned battleships has been employed as a detector shield because it is not contaminated with radioactive fallout.

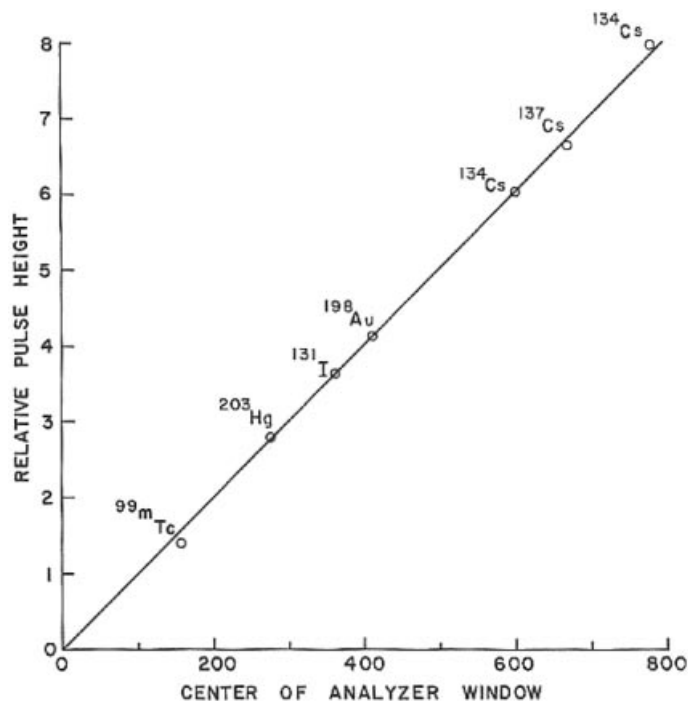


FIGURE 9-5

The average size of pulses transmitted by a pulse height analyzer as a function of the position of the center of the analyzer window.

device is plotted as a function of the position of the lower discriminator or center of the analyzer window (Figure 9-5). The maximum departure from a straight line is termed the *integral nonlinearity* and is less than 1% in a satisfactory counting system. Zero offset describes the positive or negative displacement from the origin of a curve such as that in Figure 9-5.

Multichannel Analyzers

To determine the number of pulses of different sizes impinging upon a single-channel pulse height analyzer, counts must be recorded for selected intervals of time while the window of the analyzer is moved incrementally from the smallest to the largest pulse size encountered. This procedure is tedious and imprecise because each pulse height “channel” is sampled independently and for only a short time. Also, pulse height spectra for isotopes with short half-lives are difficult to measure with a single-channel analyzer. With a single-channel analyzer and a counting time of 1 minute per channel, more than 100 minutes are required to sample a pulse height distribution that is divided into 100 parts. With a 100-channel analyzer, the same data may be collected in 1 minute. Multichannel analyzers are available commercially with thousands of channels.

In the multichannel analyzer, a pulse from the amplifier is fed to an analog-to-digital converter (ADC). Within the ADC, charge is stored in a capacitor in proportion to the amplitude of the incoming pulse. After receipt of the pulse, the capacitor is discharged while an oscillator emits pulses at a constant rate. The number of pulses emitted by the oscillator reflects the amplitude of the original pulse fed to the ADC. The number of oscillator pulses determines the location in the computer memory where a binary number is increased by 1 to reflect receipt by the ADC of a pulse of specified magnitude. Each storage location in the computer memory corresponds to a specific pulse amplitude, and the binary number stored at each location reflects the number of pulses of a specific amplitude received by the ADC during the counting period. These numbers can be displayed graphically on a display device to portray

the pulse height spectrum for the radionuclide. By arithmetic manipulation of data stored in the multichannel analyzer, the pulse height spectrum can be corrected for background counts, the presence of more than one radionuclide in the sample, and other influences.

Scalers and Timers

A scaler is used to record and display the number of pulses received from an amplifier or pulse height analyzer. Many scalars are equipped with an electronic timer to indicate elapsed counting time or to stop the accumulation of counts by the scaler after a preset counting time. Timers consist of a scaler pulsed by a constant-frequency oscillator. Elapsed or preset time may be displayed visually or printed automatically.

High-Voltage Supplies

The specifications for a high-voltage supply for a counting system vary with the detector used. A high-voltage supply for a G-M tube or a semiconductor detector need not be as stable as that for a scintillation detector, for example, because the signal from a G-M or semiconductor detector is affected little by fluctuations in applied voltage. On the other hand, the signal from a scintillation detector may vary by as much as 10% for a 1% change in high voltage. The stability of a high-voltage supply may be affected by changes in temperature, fluctuations in line voltage, or variations in the impedance (load) across the output terminals.

DETERMINE ERRORS IN RADIOACTIVITY MEASUREMENTS

The count rate measured for a radioactive sample reflects the rate of decay of atoms in the sample. However, the influence of several determinate errors must be known before the activity of the sample can be determined from the measured count rate. This influence must be included in an expression for the relationship between sample activity A and measured count rate c_{s+b} [Eq. (9-1)]:

$$A = \frac{C_{s+b}}{\{[1 - (c)_{s+b}]\tau\} EfGBf_0f_wf_s} - (c)_b \quad (9-1)$$

where corrections to the background count rate are assumed to be negligible.

- $(c)_{s+b}$ = Sample-plus-background count rate in counts per minute (cpm)
- $(c)_b$ = Background count rate (cpm)
- τ = Resolving time in minutes
- E = Detector efficiency
- f = Fractional emission of source
- G = Geometry correction
- B = Backscatter correction
- f_0 = Sidescatter correction
- f_w = Correction for attenuation in detector window, air, sample covering, etc.
- f_s = Correction for self-absorption

These corrections are discussed in the following sections.

Background Count Rate

A counting system used to measure radiation from a radioactive sample almost always indicates the presence of small amounts of radiation when the radioactive sample is removed. The residual radiation (background) originates from a number of sources,

Background radiation contributes an annual whole-body radiation dose to each person on earth. At sea level, the background dose-equivalent rate is approximately 3.6 mSv/yr (360 mrem/yr), including about 2 mSv/yr (200 mrem/yr) from radon. The latter value is highly variable from one location to another.

When compared with other common radiation detectors, the resolving time of a G-M detector is rather long (100–300 μ s), and the resolving time of a NaI(Tl) scintillation detector is rather short (0.5–10 μ s).

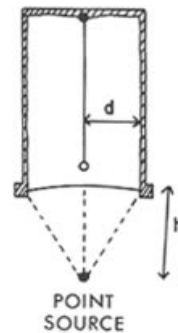
The intense radiation belts (Van Allen belts) encircling the earth were discovered because of paralysis of radiation detectors on an early unmanned space craft.

In some texts the resolving time is called the dead time, and the coincidence loss is termed the *dead-time loss*.

The detector efficiency is also termed the *intrinsic efficiency*.

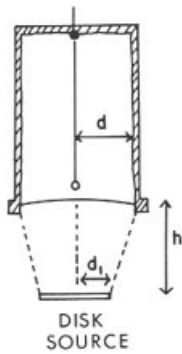
When a pulse-height window is used, the number of counts is further reduced by the fraction of counts actually transmitted by the pulse-height analyzer.

Well counters, in which the source is almost entirely surrounded by the detector, approach the condition of 4π geometry.

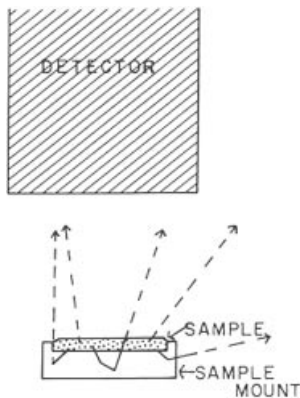


MARGIN FIGURE 9-3

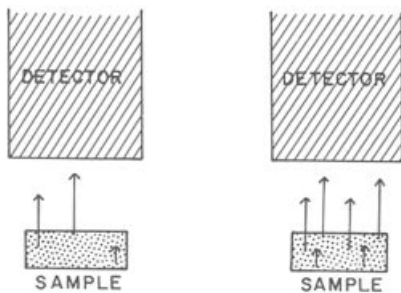
Geometric relationship between a detector and a point source of radioactive material.

**MARGIN FIGURE 9-4**

Geometric relationship between a detector and a disk source of radioactive material.

**MARGIN FIGURE 9-5**

Backscattering of radiation.

**MARGIN FIGURE 9-6**

Count rate for a sample with constant volume but increasing specific activity.

including (1) cosmic radiation, (2) radioactive materials such as ^{226}Ra , ^{14}C , and ^{40}K in the earth, human body, walls of buildings, etc., (3) radioactive materials stored or used nearby, (4) radioactive materials in devices such as watches and instrument dials, (5) radioactive contamination of the counting equipment, laboratory benches, etc., and (6) radioactive fallout (this source contributes no more than 1% to 2% to the total background radiation). Usually radiation detectors are surrounded by lead or other shielding material to reduce the background count rate. Other methods to reduce background include the use of pulse height analyzers and coincidence or anticoincidence circuits.

Resolving Time

The resolving time of a radiation detector is the interval of time required between successive interactions in the detector for the interactions to be recorded as independent events. The resolving time of a detector may be measured with an oscilloscope that displays pulses from the detector. Sequential measurements of the count rate for a short-lived radioactive nuclide may also be used to determine resolving time.²

Detectors, and the counting systems in which they are used, may be classified as one of two types:

- *Nonparalyzable systems* in which, if an interaction occurs within the resolving time of a preceding event, the second interaction is not registered but has no effect on subsequent events. Most electronic components, with the exception of radiation detectors, exhibit nonparalyzable behavior.
- *Paralyzable systems* in which the second interaction, even though it is not registered in computer memory, extends the resolving time beyond that for the first event. Most radiation detectors exhibit paralyzable behavior, a phenomenon referred to as “pulse pile-up.”

Observed versus true count rates are depicted in Figure 9-6 for nonparalyzable and paralyzable counting systems.

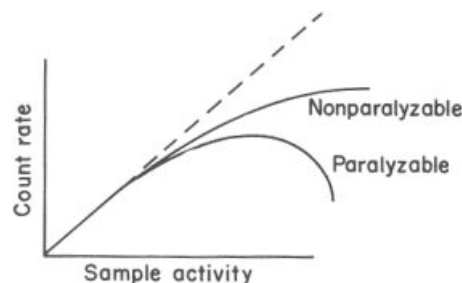
For nonparalyzable systems, the count rate c_0 corrected for the loss of counts caused by resolving time is

$$c_0 = \frac{c}{1 - c\tau} \quad (9-2)$$

where c is a measured count rate in counts per minute and τ is the resolving time in minutes. The coincidence loss is the difference between the uncorrected count rate c and the corrected count rate c_0 .

Detector Efficiency

The efficiency of a radiation detector is the quotient of the number of particles or photons that interact in the detector divided by the number of particles or photons

**FIGURE 9-6**

Count rate characteristics of paralyzable and nonparalyzable counting systems.

that enter the detector. The efficiency of most gas-filled detectors is close to 1 for α - and β -particles and about 0.01 for x and γ rays. The efficiency of a NaI(Tl) crystal or semiconductor detector for the detection of x and γ rays varies with the thickness of the crystal and energy of the x and γ rays.

Fractional Emission

The fractional emission of a radioactive sample is the fractional number of decays that yield the radiation to be detected. For example, suppose that γ rays (0.393 MeV) from ^{113m}In are detected. Because the coefficient of internal conversion is 0.55 for ^{113m}In , the number of decays that furnish a γ ray of 0.393 MeV is $1/(1 + 0.55) = 0.645$. Hence the fractional emission is 0.645 for the 0.393 MeV γ rays from ^{113m}In .

Detector Geometry

The *geometry correction* is the ratio of the number of particles or photons emitted in the direction of a detector to the total number of particles or photons emitted by the sample. Usually, the detector is said to “subtend” a certain solid angle Ω measured in steradians. A sphere subtends a solid angle of 4π steradians, and the geometry correction G is

$$G = \frac{\Omega}{4\pi}$$

The expressions 2π geometry and 4π geometry are used to describe counting conditions where half of all of the radiation emitted by the source is intercepted by the detector.

The geometry correction varies with the radius d of the detector and with the distance h between the detector and the source. The correction is described in Eq. (9-3) for point and disk sources of radioactive material:

Point source:

$$G = 0.5 \left(1 - \frac{h}{\sqrt{h^2 + d^2}} \right) \quad (9-3)$$

Disk source (radius of disk = d_1):

$$G = 0.5 \left(1 - \frac{h}{\sqrt{h^2 + d^2}} \right) - \frac{3}{16} \left(\frac{dd_1}{h^2} \right)^2 \left(\frac{h}{\sqrt{h^2 + d^2}} \right)$$

Scattering

Radiation emitted from a radioactive source in a direction outside the solid angle subtended by a detector may be scattered toward the detector as the radiation interacts with the sample container or with shielding around the sample and detector. The scattered particles or photons are backscattered if they were originally emitted in a direction away from the detector (see margin illustration). If the backscattered radiation originates entirely within the mount for the sample, the percent backscatter (%B) and backscatter factor (B) are

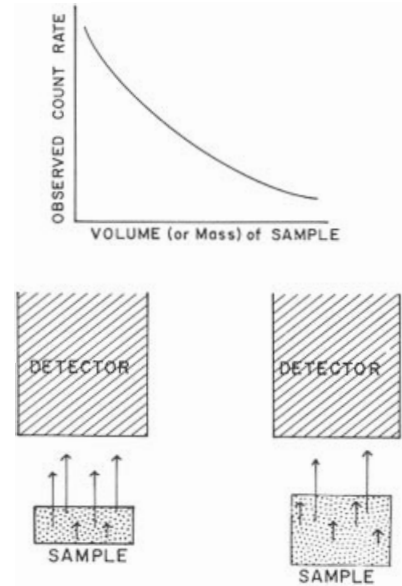
$$\%B = \left[\frac{\text{Counts with mount present} - \text{Counts with mount removed}}{\text{Counts with mount removed}} \right] \times 100 \quad (9-4)$$

$$B = \frac{\text{Counts with mount present}}{\text{Counts with mount removed}} \quad (9-5)$$

The backscatter factor is related to the percent backscatter by

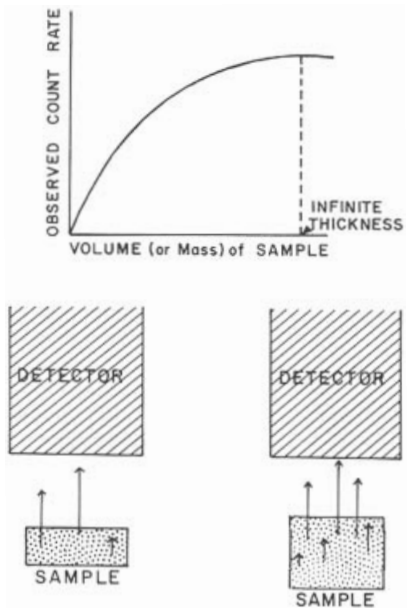
$$\%B = 100(B - 1)$$

For β -particles, the backscatter factor increases with the atomic number of the sample mount and, initially, with the thickness of the mount. *Saturation thickness* is



MARGIN FIGURE 9-7

Count rate for a sample with constant activity but increasing sample volume.

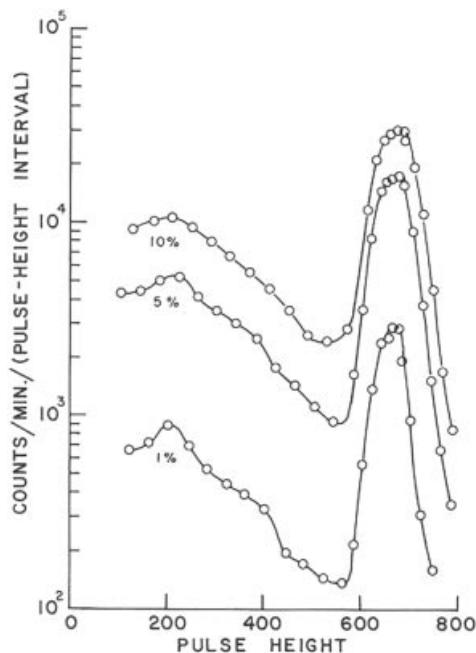


MARGIN FIGURE 9-8

Count rate for a sample with constant specific activity but increasing volume (and consequently, increasing total activity).

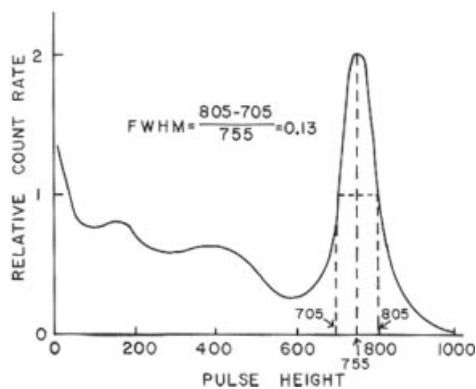
Scintillations created in solid substances by ionizing radiation were first observed by Sir William Crookes in 1903.

The study of pulse height spectra is termed *pulse height spectroscopy*, sometimes referred to as *scintillation spectroscopy*.



MARGIN FIGURE 9-9

Increased count and decreased resolution (depicted by a widening of the photopeak) are achieved as the window width is increased. Window widths are expressed as a percentage of the highest setting of the pulse height scale. Spectra were measured for a ^{137}Cs source in a 2×2 -in. NaI(Tl) well detector.



MARGIN FIGURE 9-10

“Full width at half-maximum.”

achieved when an increase in thickness does not increase the backscattered radiation. For β -particles, saturation thickness equals about three-tenths of the range of the particles in the mount. Backscattering may be reduced by using a sample mount of low atomic number. For β counting, for example, aluminum planchets are often preferred over copper or steel planchets because the backscatter factor is lower with aluminum. If the sample mount provides saturation thickness, then the backscatter factor for β -particles is virtually independent of the energy of the β -particles.

Sidescatter may be reduced by moving all scattering material (e.g., shielding around the source and detector) far from the path between the source and the detector. Sidescatter may also be reduced with a sleeve of plastic, aluminum, or other low-Z material inside the high-Z shield around the source and detector.

Air and Window Absorption

Radiation is attenuated by material between the radioactive source and the sensitive volume of the detector. Usually, at least three different attenuators are encountered by radiation moving toward the detector. These attenuators are (1) the covering over the radioactive source, (2) air between the source and the detector, and (3) the entrance window of the detector.

Self-Absorption

The term *self-absorption* refers to the attenuation of radiation within the radioactive sample itself. Shown in the margin is the increase in count rate for a radioactive sample achieved if the activity of the sample is increased with no change in sample volume. The data describe a straight line because the fraction of radiation absorbed by the sample remains unchanged. However, if the volume of the sample increases with no change in activity, then the count rate for the sample decreases because an increasing fraction of the radiation is absorbed. For a sample with increasing volume and constant specific activity (specific activity equals the activity of the sample divided by the mass of the sample), the count rate approaches a constant value at large thickness (“infinite thickness”) because the increased absorption of radiation by the sample compensates for the increased activity.

■ GAMMA-RAY SPECTROMETRY

Attenuation coefficients for photoelectric, Compton and pair-production interactions of γ rays in NaI(Tl) are plotted in Figure 9-7 as a function of γ -ray energy. For γ rays with energy less than about 200 keV, photoelectric absorption is the most probable interaction in NaI(Tl). Compton scattering is the dominant interaction between about 200 keV and 7 MeV, and pair production is most important at energies greater than 7 MeV. A γ ray that interacts in a NaI(Tl) crystal may deposit part or all of its energy in the crystal. The size of the resulting voltage pulse from the photomultiplier tube coupled to the crystal is proportional to the energy deposited in the crystal by the γ ray. Pulses from a scintillation detector may be sorted with a scintillation spectrometer to yield a pulse height spectrum that reflects the energy distribution of the interactions of incident photons.^{3,4}

■ PULSE HEIGHT SPECTRA

The pulse height spectrum for a radioactive source consists of peaks and valleys that reveal the energy of the γ rays from the source. The spectrum also reflects interactions that occur before the photons reach the scintillation detector, interactions in the detector, and the operating characteristics of components of the counting system. If the amplifier gain or the high voltage across the photomultiplier tube is increased,

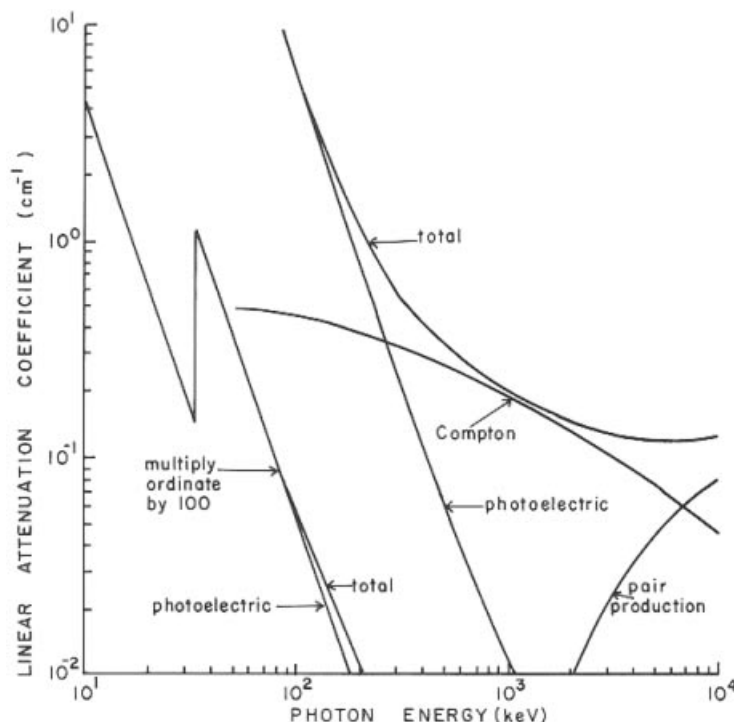


FIGURE 9-7

Linear attenuation coefficients of NaI(Tl) for photoelectric absorption, Compton scattering, and pair production as a function of γ -ray energy.

then each pulse is amplified proportionally and displayed at a position farther along the pulse height scale. Hence an increase in high voltage or gain expands the pulse height spectrum over a greater range of pulse heights (Figure 9-8). The height of the spectrum decreases because the voltage pulses encompass a wider range of pulse heights and each pulse height interval contains fewer pulses.

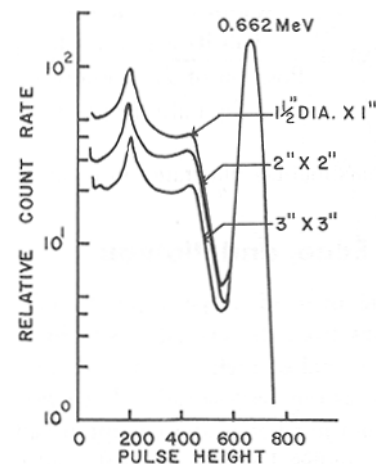
As the window of the pulse height analyzer is widened, more pulses are transmitted to the display device for each position of the window along the pulse height scale. Consequently, the count rate at each window position increases with window width. With a wider window, however, peaks and valleys in the spectrum are defined less precisely, and the resolution of the spectrum is reduced. Pulse height spectra for ^{137}Cs measured with analyzer windows of different widths are shown in Margin Figure 9-9.

Photopeaks

A *photopeak* on a pulse height spectrum encloses pulses produced by total absorption of γ rays of a particular energy in the scintillation crystal. The energy of the γ ray may have been deposited in the crystal during one interaction or a series of interactions. The *photofraction* or *peak-to-total ratio* for a photopeak is the area enclosed within the photopeak divided by the total area enclosed within the pulse height spectrum. The intrinsic peak efficiency is the fraction of the γ rays of a particular energy impinging upon a detector that produces pulses enclosed within the photopeak. The *resolution* of a pulse height spectrum is sometimes described as the *full width at half-maximum* (FWHM), where

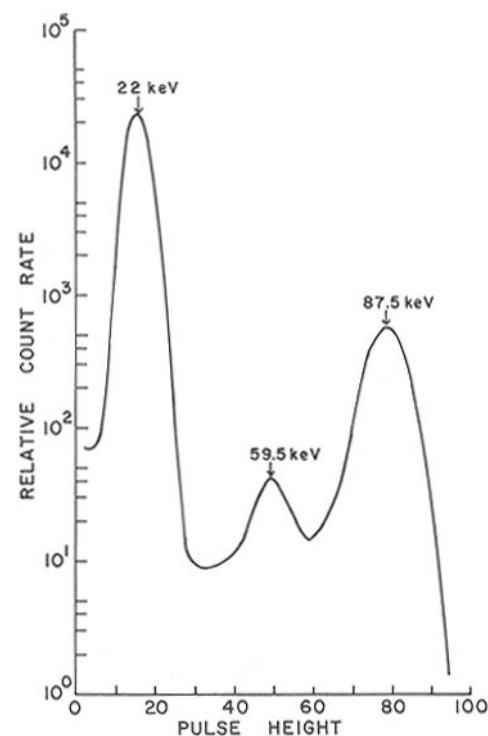
$$\text{FWHM} = \frac{\text{Width of the photopeak at half its maximum height}}{\text{Position of the photopeak along the pulse height scale}} \quad (9-6)$$

The full width at half-maximum is illustrated in Margin Figure 9-10.



MARGIN FIGURE 9-11

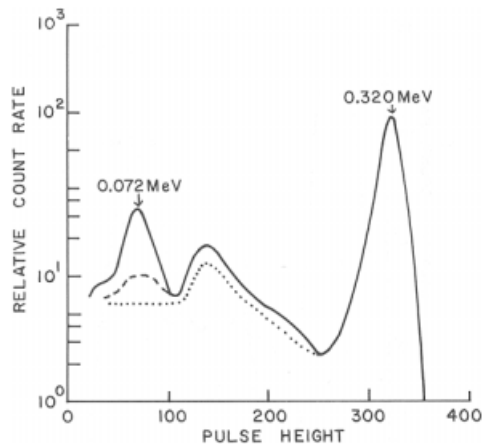
Pulse height spectra for ^{137}Cs measured with NaI(Tl) crystals of different sizes. The photopeaks are normalized to the same height. (From Heath, R.³ Used with permission.)



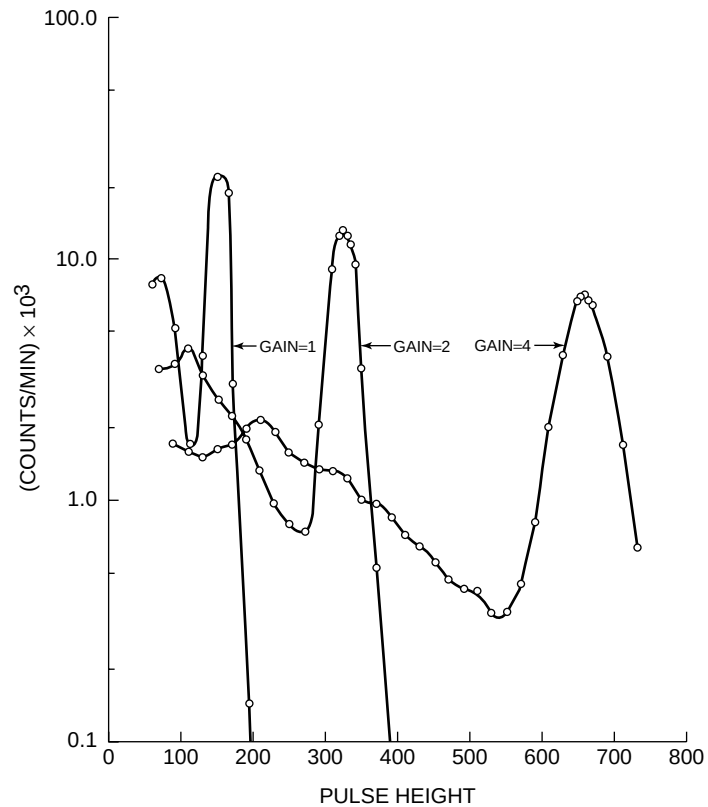
MARGIN FIGURE 9-12

Pulse height spectrum for ^{109}Cd . An iodine x-ray escape peak is present at a position 28 keV below the photopeak for the 87.5-keV γ rays from ^{109}Cd . The large photopeak at 22 keV is produced by absorption of characteristic x rays released during decay of ^{109}Cd by electron capture. (From O'Kelley, G.⁴ Used with permission.)

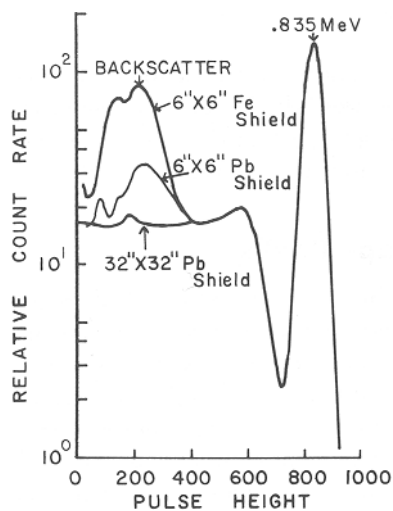
The x-ray escape peak is occasionally referred to as an "iodine escape peak" because the peak reflects escape of an x ray produced by an L $\rightarrow$ K electron transition in iodine.

**MARGIN FIGURE 9-13**

Characteristic x-ray peak at 72 keV produced by x-rays from a 6 × 6-in. lead shield surrounding a scintillation detector and a ^{51}Cr source. The x-ray peak is reduced by increasing the inner dimensions of the shield to 32 × 32 in. (dotted curve) or by lining the shield with 0.03 in. of cadmium (dashed curve). (From Heath, R.³ Used with permission.)

**FIGURE 9-8**

Pulse height spectra for ^{137}Cs measured with different settings of the amplifier gain.

**MARGIN FIGURE 9-14**

The backscatter peak at about 200 keV is produced by absorption in the detector of photons scattered at wide angles during Compton interactions of primary γ rays in the detector-source shield. The backscatter peak is greatest for the 6 × 6-in. iron shield because the relative probability of Compton interaction of the 0.835-MeV γ rays from ^{54}Mn is higher for iron than for lead. The backscatter peak is reduced if the distance between the shield and the detector is increased.

Compton Valley, Edge, and Plateau

Compton interactions of incident γ rays in a NaI(Tl) crystal, with escape of the scattered photons from the crystal, result in the formation of voltage pulses smaller than those enclosed within a photopeak. The valley just below a photopeak is referred to as the *Compton valley*. The *Compton edge* is the portion of the pulse height spectrum just below the Compton valley where the height of the spectrum increases rapidly. The portion of the pulse height spectrum below the Compton edge is termed the *Compton plateau*.

The relative heights of the photopeak and the Compton plateau vary with the size of the NaI(Tl) crystal. In a larger crystal, more scattered photons interact before they escape from the crystal. Hence the height of the photopeak is increased, and the height of the Compton plateau is reduced (see Margin Figure 9-11).

X-Ray Escape Peaks

A γ ray that interacts photoelectrically in a NaI(Tl) crystal usually ejects an electron from the K shell of iodine. As the vacancy left by the photoelectron is filled by an electron from the L shell, a characteristic x ray is released with an energy equal to the difference (28 keV) in binding energy of L and K electrons in iodine. If the 28-keV x ray interacts in the scintillation crystal, then the light released during this interaction contributes to formation of the voltage pulse for the primary γ ray, and this voltage pulse falls within the photopeak. However, if the x ray escapes from the crystal, then the pulse for the primary γ ray is smaller than those enclosed within the photopeak. If many x rays escape during γ ray interactions, then a peak occurs on the pulse height spectrum at a location about 28 keV less than the photopeak (Margin Figure 9-12). This peak is referred to as an *x-ray escape peak*. X-ray escape peaks are not observed for

γ rays with energy greater than about 200 keV because the lateral spread of the photopeak for photons of higher energy obscures the presence of the x-ray escape peak. Also, characteristic x rays released during interactions of higher-energy γ rays have a lower probability of escape from the crystal because more energetic γ rays tend to penetrate farther into the crystal before interacting.

Characteristic X-Ray Peak

Characteristic x rays are released as γ rays from a radioactive source undergo photoelectric interaction in the lead shield surrounding the source and detector. Some of these x rays escape from the shield and strike the NaI(Tl) crystal to produce peaks in the pulse height spectrum. In Margin Figure 9-13, the peak at 0.072 MeV reflects the absorption of characteristic x rays from a lead shield surrounding a scintillation detector and a ^{51}Cr source. The number of characteristic x rays reaching the detector may be reduced by increasing the distance between the detector and shield or by lining the shield with a material with atomic number less than that of lead.

Backscatter Peak

If a photon with energy greater than 200 keV is scattered at a wide angle during a Compton interaction, then the scattered photon has an energy of about 200 keV irrespective of the energy of the primary γ ray. For this reason, photons scattered at wide angles during Compton interactions in the detector-source shield produce a peak at about 200 keV on the pulse height spectrum (see Margin Figure 9-14). This peak is referred to as the *backscatter peak*. The number of scattered photons received by the detector may be reduced by increasing the distance between the shield and the detector or by choosing a material for the shield in which the number of Compton interactions is reduced.

Annihilation Peak

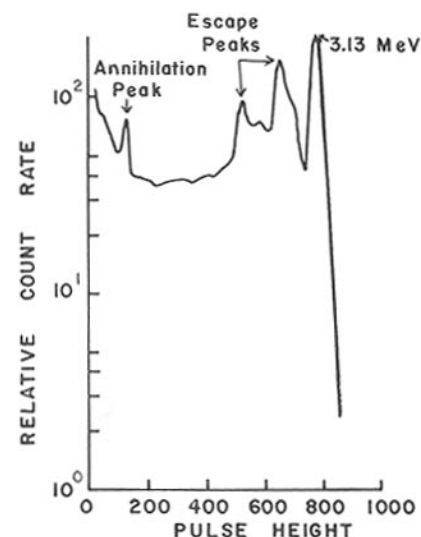
Pair-production interaction of γ rays with an energy greater than 1.02 MeV is accompanied by release of 511-keV annihilation photons. If a primary γ ray interacts by pair production in the detector-source shield, then one of the annihilation photons may escape from the shield and interact in the crystal. This process results in the production of an annihilation peak of 511 keV in the pulse height spectrum (see margin). An annihilation peak at 511 keV is also present in pulse height spectra for positron-emitting nuclides because annihilation radiation is released as the positron combines with an electron.

Annihilation Escape Peaks

Pair production of high-energy (>1.02 MeV) photons in the scintillation crystal results in the emission of two 511-keV annihilation photons, one or both of which may escape from the crystal. Pulses that reflect the loss of one annihilation photon contribute to the *single-escape peak* that occurs at a location 0.511 MeV below the photopeak for the primary γ ray. Pulses that reflect the loss of both annihilation photons contribute to the *double-escape peak* that occurs at a location 1.02 MeV below the photopeak. Single- and double-escape peaks, together with an annihilation peak at 0.511 MeV, are present in the pulse height spectrum for ^{37}S , a nuclide that decays by the emission of a β -particle and a γ ray with an energy of 3.13 MeV (Margin Figure 9-15).

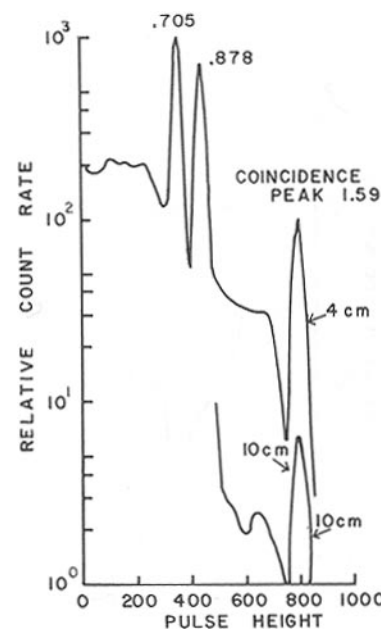
Coincidence Peak

Many radioactive sources decay with the emission of two or more γ rays in cascade. When γ rays from these sources are detected, more than one of the cascading γ rays may be completely absorbed in the crystal. Coincidence peaks in the pulse height



MARGIN FIGURE 9-15

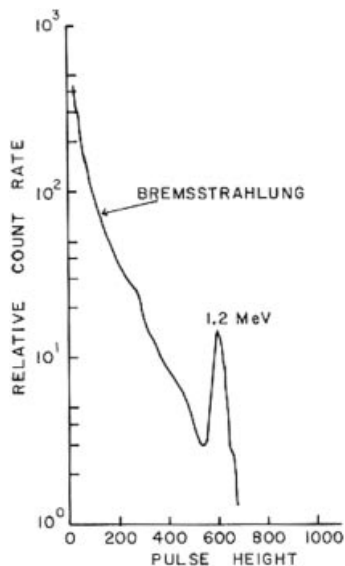
The pulse height spectrum for ^{37}S illustrates single and double escape peaks and an annihilation peak at 0.511 MeV.



MARGIN FIGURE 9-16

The coincidence peak reflects the simultaneous absorption of 0.705-MeV and 0.878-MeV γ rays from ^{94}Nb . The relative height of the coincidence peak is reduced if the distance between source and detector is increased from 4 cm to 10 cm.

Coincidence peaks are often termed “sum peaks.”



MARGIN FIGURE 9-17

The pulse height spectrum of ^{91}Y illustrates the bremsstrahlung contribution characteristic of a radioactive source for which the ratio of γ rays to β -particles is low.

The fraction of detected γ rays that contribute to the photopeak is known as the *photofraction* or *photopeak efficiency*.

The width of the photopeak is a reflection of statistical variations that occur in the process of forming voltage pulses in the detector. These variations include:

- Variations in the number of light photons released in the crystal per unit energy absorbed
- Variations in the number of light photons that must impinge on the photocathode to eject a photoelectron
- Variations in the number of electrons released at each successive dynode in the pm tube.

spectrum reflect the simultaneous absorption of more than one γ ray in the crystal (Margin Figure 9-16). The relative height of a coincidence peak is reduced if the distance is increased between source and detector, because the probability is reduced that two or more photons will strike the crystal simultaneously. For high-activity samples, coincidence peaks may be produced by simultaneous absorption of γ rays released during the decay of different atoms of the sample.

Bremsstrahlung

Bremsstrahlung is released as β -particles from a radioactive source are absorbed in the source or source-detector shield. If the ratio of γ rays to β -particles emitted by the source is low, then the contribution of bremsstrahlung to the pulse height distribution may be noticeable, particularly at the lower end of the pulse height scale (Margin Figure 9-17).

PHOTOPEAK COUNTING

Radioactive sources that emit γ rays are often counted by sending to the display device only those pulses that are enclosed within a pulse height window centered over the photopeak. Other pulses are rejected because they are larger or smaller than the pulses transmitted by the window. To center the window over the photopeak, the window is positioned at a desired location along the pulse height scale. Then the amplifier gain or the high voltage is increased until a maximum count rate is obtained, indicating that the photopeak is centered in the window. For example, if the photopeak for ^{137}Cs is desired at 662 on the pulse height scale, then the window is centered at 662, and the gain or high voltage is increased until a maximum count rate is obtained. If the size of the voltage pulse (i.e., the pulse height) varies linearly with energy deposited in the crystal, then this procedure furnishes a pulse height scale calibrated from 0 to 1000 keV, with each division on the pulse height scale representing 1 keV. The pulse height scale may be calibrated from 0 to 2000 keV by centering the ^{137}Cs photopeak at 331 and from 0 to 500 keV by centering the photopeak for a lower-energy γ ray at an appropriate position on the pulse height scale. A pulse height scale may be calibrated from 0 to 500 keV, for example, by centering the photopeak for $^{99\text{m}}\text{Tc}$ (primary γ ray of 140 keV) at 280 on the pulse height scale.

The width of the analyzer window most desirable for photopeak counting varies with the relative heights of the photopeak and background at the position of the window. For many counting situations, the background pulse-height spectrum has relatively little structure and is much lower than the photopeak for the sample. For these situations, a relatively narrow (e.g., 2%) window should be centered over the photopeak. If the photopeak is not much higher than background, then the value of the sample net count rate squared divided by the background count rate may be determined as a function of window width. When the ratio (known as the square of the signal divided by the background, or S^2/B ratio) is maximum, the fractional standard deviation is minimum for the net count rate of the sample.⁵ A plot of S^2/B as a function of window width is shown in Figure 9-9 for a ^{137}Cs source and a 2×2 -in. NaI(Tl) well detector. The optimum window is 70 keV.

Ross and Harris⁶ have shown that an acceptable window usually is achieved if lower and upper discriminators are positioned on either side of the photopeak at a location where the count rate is equal to one-third of the count rate at the center of the photopeak. About 85% of all pulses enclosed within the photopeak are transmitted by a window defined by discriminators in these positions, and the window is termed an *85% window*. The window width should be increased if the window or amplifier gain drifts or if the width of the window fluctuates. Under these circumstances the lower and upper discriminators should intersect the pulse height spectrum at equal heights near the base of the photopeak. If the gain of the amplifier is unstable, then the photopeak should be intercepted by the lower and upper discriminator at relative

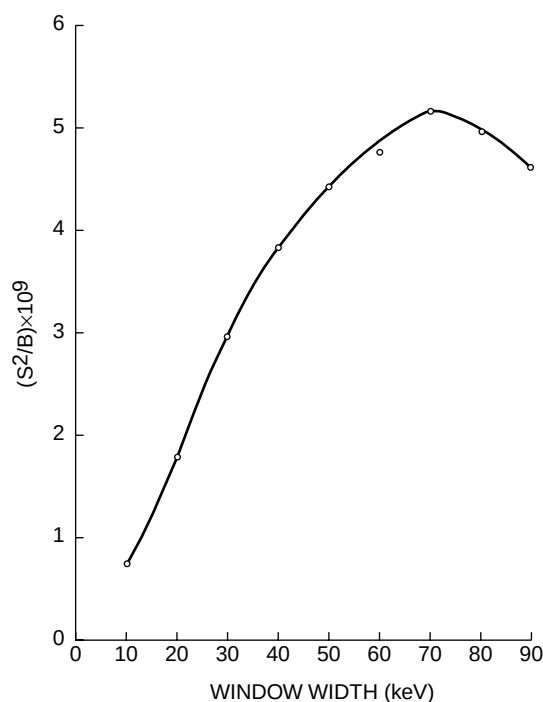


FIGURE 9-9
 S^2/B versus window width for a ^{137}Cs source and a 2×2 -in. NaI(Tl) well detector.

count rates R_1 and R_2 , respectively, where R_1 and R_2 are determined by the expression $R_1/R_2 = V_2/V_1$ and V_1 and V_2 are the positions of the lower and upper discriminators along the pulse height scale.

RADIOACTIVE AGENTS FOR CLINICAL STUDIES

Many radioactive nuclides have been administered to patients for the diagnosis and treatment of a variety of conditions. For a nuclide to be acceptable for most diagnostic applications, the half-life for radioactive decay should be between 1 hour and a few days. Nuclides with half-lives shorter than 1 hour decay too rapidly for the labeling and sterilizing of radioactive compounds, and nuclides with long half-lives often deliver high radiation doses to patients.

Most radioactive agents administered to patients are injected intravenously and must be sterile. Agents are sterilized by autoclaving (e.g., heating for 15 to 20 minutes at a temperature of about 125°C and a pressure of roughly 20 psi) or by filtration through fine cellulose-ester membranes. The injected material should have a pH near 7 and should either dissolve in an aqueous medium or form a colloidal suspension or a suspension of larger particles. An agent injected intravenously must be free of *pyrogens*, particles that have a diameter of 0.05 to $1\ \mu\text{m}$ and do not decompose at elevated temperature.⁷ The presence of pyrogens in an agent administered intravenously to a patient causes the rapid appearance of characteristic symptoms that have been documented by Miller and Tisdal.⁸ Solutions are free from pyrogens only if they are prepared by meticulous chemical techniques. Procedures for testing for the presence of pyrogens are described in the *United States Pharmacopoeia*⁹ and other publications.¹⁰

Other important considerations for a radioactive pharmaceutical to be used clinically are:

1. *Radiochemical purity*, the ratio of the quantity of the desired isotope (or nuclide) present in the desired chemical form to the total quantity of the isotope (or nuclide) in the pharmaceutical.

Optimum characteristics of a radionuclide for nuclear medicine imaging include:

- A suitable physical half-life
- Emission of γ rays of suitable energy
 - High enough to penetrate the body
 - Low enough to be absorbed by the detector collimator
- Minimum particulate radiation
- Availability in a convenient, economic and uncontaminated form

An exception to the half-life rule are the radioactive nuclides (^{11}C , ^{13}N , and ^{15}O) with extremely short half-lives (on the order of seconds or minutes) used for positron emission tomography (PET). Use of these nuclides requires a source such as a cyclotron adjacent to the PET imaging area.

The uptake ratio of a pharmaceutical in an organ is a measure of the difference in the concentration of activity in the organ (target) compared with surrounding tissues (nontarget).

Radioiodinated human serum albumin (RISA) is 99.9% radiochemically pure when freshly prepared. Over time, some of the radioactive iodine detaches from the RISA molecule, and the radiochemical purity declines. The rate of decline depends strongly on storage conditions.

An example of a problem with radionuclidic purity is ^{123}I which invariably is contaminated with ^{124}I . The ^{124}I does not contribute to a study with ^{123}I , but it does increase the radiation dose to the patient and degrades the quality of the ^{123}I image.

A carrier is a nonradioactive isotope of a particular element that is present in the pharmaceutical.

Radiolysis is the alteration of chemical structure caused by exposure to radiation. Compounds tagged with long-live radioactive nuclides and stored for significant periods of time can experience substantial radiolysis. This problem can be reduced by storing the compound as follows:

- In a diluted or dispersed form
- At a low temperature
- With free-radical scavengers added to the solution
- At reduced oxygen concentration
- Under conditions that provide maximum chemical stability

Mechanisms of Radiopharmaceutical

Localization¹¹

Mechanism	Example
Active transport	Thyroid uptake and scanning with iodine
Compartmental localization	Blood pool scanning with human serum albumin, plasma, or red blood cell volume determinations
Simple exchange or diffusion	Bone scanning with ^{99m} Tc-labeled phosphate compounds
Phagocytosis	Liver, spleen, and bone marrow scanning with radiocolloids
Capillary blockade	Lung scanning with macroaggregates (8 to 75 μ m); organ perfusion studies with intraarterial injection of macroaggregates
Cell sequestration	Spleen scanning with damaged red blood cells
Receptor binding/antibody-antigen	Tumor imaging with somatostatin, ¹¹¹ In-pentetreotide, ¹¹¹ In-OncoScint for tumors

2. *Radioisotopic purity* (or *radionuclidic purity*), the ratio of the activity of the desired radioactive isotope (or nuclide) to the total activity of radioactive material in the pharmaceutical. Progeny nuclides are usually excluded from the description of radionuclidic purity.
3. Presence of a *carrier* in the pharmaceutical. A pharmaceutical is *carrier free* if all isotopes of a particular element in the pharmaceutical are radioactive.
4. Presence of an *isotope effect*, which is an expression used to describe changes in chemical behavior of the pharmaceutical that are caused by differences in isotopic mass between the radioactive nuclide and its stable counterpart. An isotope effect is frequently observed with compounds labeled with tritium. Occurrence of the isotope effect decreases rapidly with increasing atomic number of the radioactive nuclide and is rare with compounds labeled with radioactive nuclides with atomic numbers greater than about 15.
5. Presence of *isotope exchange reactions* during which the radioactive nuclide detaches from the pharmaceutical and adheres to another compound in the body.
6. Radiation dose delivered to critical organs and to the whole body of the patient.
7. Presence of degradation products caused by chemical instability and by self-irradiation of the pharmaceutical.

Localization of a radioactive pharmaceutical administered intravenously, as well as the rate and mode of excretion of the pharmaceutical from the body, is determined by a number of properties, including^{12,13}:

1. The oxidation (valence) state of the pharmaceutical labeled with the nuclide at the pH (7.4) of blood.
2. The solubility of the labeled pharmaceutical in an aqueous medium. If the compound is insoluble, then the localization and excretion of the compound are determined by the size of the particles formed.
3. The tendency of the pharmaceutical to incorporate into or bind to organic compounds.

During most clinical applications of radioactive nuclides, radiation emitted by a nuclide distributed within the body is measured with a detector positioned outside the body. For these applications, the nuclide must emit a γ ray with an energy between about 20 and 700 keV. Photons with energy less than 20 keV are severely attenuated by the patient, and photons with energy greater than 700 keV are attenuated inadequately by the collimator attached to the detector. Nuclides (e.g., ^{99m}Tc) that decay by isomeric transition are preferred for *in vivo* studies, because no particulate radiation is emitted from the nucleus during the decay of these nuclides and, consequently, the radiation dose delivered to the patient is reduced.

PROBLEMS

*9-1. A G-M detector that is 60% efficient for counting β -particles subtends a solid angle of 0.2 steradians around a sample of ²⁰⁴Tl ($T_{1/2} = 3.8$ years). The counter registers 2 cps. Assuming that corrections for backscatter, sidescatter, air, window, and self-absorption all equal 1.0, compute the number of ²⁰⁴Tl atoms in the sample.

*9-2. A G-M detector has a resolving time of 300 μ s. What is the count rate corrected for coincidence loss if the measured count rate is 10,000 cpm? What is the maximum count rate if the coincidence loss must not exceed 1%?

*9-3. The following data were recorded for identical samples:

"Weightless" Mylar mount	2038 cmp
Silver backing	3258 cpm

Compute the backscatter factor and the percent backscatter.

9-4. Explain why there is a saturation or infinite thickness for curves of backscattering (count rate versus thickness of backing material)

and self-absorption (count rate versus weight of sample of constant specific activity).

- 9-5. The following data were obtained by counting a series of weighed fractions of a radioactive sample:

Mass (mg)	2	4	6	8	10	15	20	25	30	53
cpm (net)	85	160	235	305	335	360	380	398	408	514

Plot a self-absorption curve and determine the specific count rate (cpm/mg) for the sample corrected for self-absorption.

- *9-6. A ^{131}I standard solution was obtained from the National Bureau of Standards (NBS). The following information was supplied concerning the solution: assay of ^{131}I at 8 A.M., January 1, 2.09×10^4 dps/mL $\pm 2\%$. On January 15 at 8 A.M., a sample of ^{131}I was counted. This sample had been obtained from a shipment of ^{131}I in the following manner: 20 λ ($1 \lambda = 1 \mu\text{L}$) of the shipment was diluted in a volumetric flask to 25 mL; 100 λ of this solution was diluted to 10 mL; and 50 λ of this solution was withdrawn, mounted on Mylar, dried, and counted. The dried sample provided a count rate of 4280 cpm corrected for background. At the same time (8 A.M., January 15), a 50- λ sample of the NBS standard ^{131}I solution was mounted, dried, and counted in the same manner. The count rate for the standard sample was 1213 cpm corrected for background.

- What was the specific activity (millicuries per milliliter) of the ^{131}I shipment at the time the sample from the shipment was counted?
- What was the total correction factor between the count rate and the activity of the system used to count ^{131}I mounted on Mylar?

- 9-7. A series of radioactive samples of short half-life will be counted one per day over several days. Explain how variations in the observed counts caused by fluctuations in the sensitivity of the counting system can be compensated.

- 9-8. A pulse height spectrum for ^{131}I (γ ray of 0.393 MeV) was obtained with a single-channel pulse height analyzer calibrated from 0 to 500 keV. The window width was 2%. Describe the changes in the spectrum if

- A 5% window were used
- The amplifier gain were twice as great
- The amplifier gain were half as great
- The upper discriminator of the window were removed

- *9-9. ^{24}Na decays to ^{24}Mg by β decay with a $T_{1/2}$ of 15 hours. Gamma rays of 4.14 MeV (<1%), 2.76 MeV (100%), and 1.38 MeV (100%) are emitted during the decay process. A pulse height spectrum for this nuclide exhibits peaks at 2.76 MeV, 2.25 MeV, 1.74 MeV,

1.38 MeV, 0.87 MeV, 0.51 MeV, and about 0.20 MeV. Explain the origin of each of the peaks.

- 9-10. Describe the changes in a pulse height spectrum for ^{51}Cr if
- A larger NaI(Tl) crystal were used
 - A 1-in. slab of Lucite were interposed between the ^{51}Cr source and the scintillation detector
 - A 1-in. slab of Lucite were placed behind the ^{51}Cr source

- *9-11. The position of the photopeak for ^{137}Cs γ rays (0.662 MeV) changes with high voltage as shown in the following table when the gain is set at 4.

Position of photopeak on pulse height scale	520	580	650	730	820
High voltage across photomultiplier tube	730	750	770	790	810

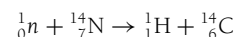
If the gain is reduced to 2 and the high voltage is set at 750, where is the photopeak for the 1.33-MeV γ ray from ^{60}Co on the pulse height scale?

- 9-12. Describe the decay characteristics desired for a radioactive nuclide used as a scanning agent.

- 9-13. Describe the characteristics desired for a chemical compound used as a scanning agent.

- *9-14. A shipment is labeled "30% ^{14}C -enriched $\text{Ba}^{14}\text{CO}_3$ —weight 2285 mg." This label is interpreted to mean "30% by weight $\text{Ba}^{14}\text{CO}_3$, the rest being $\text{Ba}^{12}\text{CO}_3$ and $\text{Ba}^{13}\text{CO}_3$." What is the activity of ^{14}C in the sample?

- *9-15. Radioactive ^{14}C is produced in the upper atmosphere by neutrons released during cosmic ray bombardment:



The radioactive carbon is distributed over the earth as $^{14}\text{CO}_2$ in the air and $\text{NaH}^{14}\text{CO}_3$ in the sea. Plants and animals incorporate ^{14}C while alive. After a plant or animal dies, the incorporated ^{14}C decays with a half-life of 5600 years. The radioactivity in the remains of the plant or animal indicates the time elapsed since the death of the organism. Determination of the age of objects by measurement of their ^{14}C content is termed *carbon dating*. For example, wood from an old mummy case provides a specific count rate of 10.7 cpm/g. New wood furnishes 16.4 cpm/g. How old is the mummy case?

- *9-16. What is the specific activity (curies per gram) of a pure sample of ^{32}P ?

*For problems marked with an asterisk, answers are provided on p. 492.

SUMMARY

- Components of a nuclear counting system include:
 - Detector
 - High voltage supply
 - Preamplifier

- Amplifier
- Pulse height analyzer
- Scaler
- Timer
- Factors contributing to determinate error in nuclear counting include:
 - Background count rate
 - Resolving time
 - Detector efficiency
 - Fractional emission of source
 - Geometry
 - Backscatter
 - Sidescatter
 - Attenuation
 - Self-absorption
- Characteristics of a pulse-height spectrum include:
 - Photopeak
 - Compton valley, edge, and plateau
 - X-ray escape peak
 - Backscatter peak
 - Characteristic x-ray peak
 - Annihilation peak
 - Annihilation escape peak
 - Coincidence peak
 - Bremsstrahlung
- Properties of radioactive pharmaceuticals include:
 - Sterility
 - Pyrogen-free
 - Radiochemical purity
 - Radionuclidic purity
 - Presence of a carrier
 - Presence of an isotopic effect
 - Presence of isotope exchange reactions
 - Radiation dose
 - Presence of degradation products
- Localization and excretion of a radioactive pharmaceutical depend on:
 - The oxidation state of the pharmaceutical
 - The solubility of the pharmaceutical
 - The incorporation or binding capacity of the pharmaceutical

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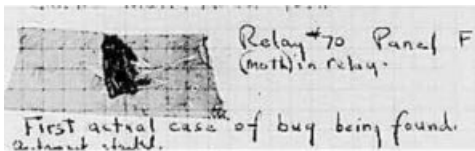
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Babbage once remarked that his calculating machine “could do everything but compose country dances.”

Moore’s Law: In 1965, Gordon Moore, co-founder of Intel Corporation, mentioned in a speech that the amount of information that can be stored in a silicon computer chip (related to the number of transistors per chip) doubles every 18–24 months. Moore’s Law, as it came to be known, held through the rest of that century. It is not expected to hold for more than another decade or so into the twenty-first century, because fundamental limitations such as the quantum uncertainty of how individual electrons behave in a circuit may be reached.



A



B

MARGIN FIGURE 10-1

Grace Hopper, Ph.D., a pioneer of electronic computing and computer language development, worked on most of the “mainframe” computers in existence from the 1940s to the 1980s. She developed many of the approaches to programming and data base management that are taken for granted today. Much of this effort was during military service, in which she rose to the rank of Admiral in the U.S. Navy. In 1951, Grace Hopper coined the term “bug” for the cause of any computer malfunction, after finding an actual bug in an electronic relay of an early computer.

OBJECTIVES

After completing this chapter, the reader should be able to:

- Convert numbers from base 10 to other bases.
- Explain how electrical circuits mimic logical operations and how that allows mathematical operations to be performed by a computer.
- Define the terms bit, byte, and word.
- Describe how letters, numbers, and images are stored by a computer.
- Compute image size and transmission time over a network.
- Calculate the error in a digitized value.
- Name the main hardware components of a computer.
- List some standard input and output devices.
- Discuss the difference between serial and parallel operations.
- Give several examples of software.
- Name the typical hardware components of a network.
- Give examples of the speed of transmission commonly found in
 - Copper wire
 - Wireless
 - Fiber optic
- State the requirements for monitor performance and matrix size found in the American College of Radiology standard.

HISTORY

A computer is any device other than the human brain that records and operates with numbers and symbols. By this definition, computers were invented by primitive humans when they counted on their fingers or piled up sticks and stones to record changes in some quantity. The modern electronic computer is fundamentally no different, except that it replaces sticks and stones with a series of on–off switches. These switches can be set and changed automatically, which makes the modern computer much more useful than a simple counting device. Moreover, multiple meanings and interrelationships have been developed among the on–off switches. This is the property that sets the modern computer apart from any other computational device or technique.

One of the earliest calculating devices was the abacus, which in its most common form consists of beads on a wire frame. There is evidence of early forms of the abacus in use by the Babylonians as early as 3000 B.C.¹

One of the first mechanical calculating devices was invented by Pascal in the 1600s.^{2,3} It was an adding machine that freed accountants from summing long chains of numbers by hand. It performed only the single operation of addition. In 1801, the automated loom was invented by Jacquard. This device used punched cards to control the operation of the loom. It could be reprogrammed to produce different weaves by changing the pattern of holes in the cards. The automated loom did not actually perform calculations; however, it introduced the concept of programming, which is essential to modern computers.

In 1833, Babbage, an English mathematician, designed an “analytical engine” to produce navigational tables for commercial shipping.⁴ The Babbage invention used decks of punched cards to program different functions. It represented the zenith of mechanical calculating devices. Although the working model of the engine was not produced during the inventor’s lifetime, advances in the technology of precision machine tooling in this century have led to construction of working models that verify the soundness of the original design.

In 1890, a new type of tabulating machine was developed by Hollerith for the U.S. Census Bureau.⁵ This device received data from punched cards, kept track of

separate categories of data, and kept an updated sum of numbers as more information was entered. It represented the first system for automated entry of information into a “data base” for the management of large amounts of information.

The first true automatic, electronic, digital computer was built in 1939 at Iowa State University by Atanasoff and Berry. This device was designed to help solve equations in quantum mechanics.⁶ Additional electronic computers that were stimulated by the technologic demands of World War II evolved a few years later. The Colossus was built in 1943 in Betchley, England, by Turing and others to penetrate German codes during World War II.⁷ The Electronic Numerical Integrator and Calculator (ENIAC) (Margin Figure 10-2) was constructed by Mauchly and Eckert at the University of Pennsylvania in 1945 to prepare ballistics tables.⁸

MACHINE REPRESENTATION OF DATA

The most fundamental unit of information is a simple yes–no decision. Components in a computer can record and store this information as a simple “current will pass–current will not pass” or “voltage is high–voltage is low” condition. Symbolically, we refer to this polarized situation with the digits 1 and 0. The binary “base 2” number system describes the use of ones and zeros to describe quantities numerically and to perform mathematical operations. The binary number system is a natural way to describe the operation of a computer or to “communicate” with one. For the same reason, number systems based upon powers of 2 such as octal (base 8) and hexadecimal (base 16) are also used in computer applications.

Numbers and Bases

The number system with which we are most familiar is the decimal, or “base 10,” system. It consists of the 10 digits 0 through 9, and it denotes a power of 10 through the position of the digit in a number. If a certain power of 10 is not needed to express a quantity, its position is occupied by a zero. These characteristics, a series of digits in a “place holder” configuration that connotes different powers of a base, are shared by most number systems including binary, octal, base 10, and hexadecimal.

The binary, or base 2, system is illustrated in Margin Figures 10-3 and 10-4. The numbers 0 and 1 are the same in both base 2 and base 10 systems. However, the decimal number 2 has no symbol in binary since only 0 and 1 are available as digits. The problem of binary expression of the number 2 is analogous to the problem of decimal expression of the number 10 (Margin Figure 10-5). Because there is no single digit in the decimal system to express the number 10, a place is marked in the rightmost column with the digit zero, and a 1 is placed in the next column to the left, the “ten’s” place, to express the decimal number “10.” Similarly, to express the number 2 in binary, a zero is positioned in the rightmost column, and a 1 is placed in the next column to the left, the “two’s” place.

Octal and hexadecimal number systems are based upon the numbers 8 and 16, respectively. The properties of these number systems are illustrated in Margin Figure 10-3. Conversion of number representations among the decimal, binary, octal, and hexadecimal systems is illustrated in Margin Figure 10-6. When working with numbers in more than one base, it is important to remember that the quantity itself—for example, “twenty things”—is the same in any number system. It is only the notation that changes when moving from one number base to another.

Numerical and Logical Operations

One may pose the question, “How do computers think?” One should then counter with the question, “How do humans think?” These questions are topics of current research interest, and it is tempting to presume that the best, or at least the most

In the 1930s the word “computer” meant a person (typically female) whose job was doing computations (Davis, M. *The Universal Computer: The Road from Leibniz to Turing*. New York, W. H. Norton, 2000.)



MARGIN FIGURE 10-2

The Electronic Numerical Integrator and Computer (ENIAC) was constructed in 1945 under military contract at the University of Pennsylvania for approximately \$500,000. It contained 17,468 vacuum tubes, weighed 60,000 pounds, occupied 16,200 cubic feet of space, and used 174 kilowatts of power. It operated for nine years, during which time it was the fastest computer on earth. It could add 5000 numbers/second, whereas a typical PC today can add 70 million/second. As part of ENIAC’s 50th Anniversary Celebration, a team of graduate students at the University of Pennsylvania duplicated its functionality on a $7.44 \times 5.29\text{-mm}^2$ chip that contains about 174,569 transistors. A typical PC central processing chip used today contains over three million transistors.

Binary (Base 2)	Decimal (Base 10)
0	0
1	1
10	2
11	3
100	4
101	5
110	6
111	7
1000	8
1001	9
1010	10

MARGIN FIGURE 10-3

Binary and decimal notation for numbers.

Claude Shannon (1917–2001) founded what is known today as “Information Science.” In a landmark 1940 paper, he wrote down the equations that specify how much information communication networks can carry, described how signals could be conveyed by a series of ones and zeros (which we now call bits), and introduced the concept of data compression.

1	2^{10}	one thousand twenty fours	1024
0	2^9	five hundred twelves	512
0	2^8	two hundred fifty sixes	256
1	2^7	one hundred twenty eights	128
0	2^6	sixty fours	64
1	2^5	thirty twos	32
1	2^4	sixteens	16
0	2^3	eights	8
0	2^2	fours	4
1	2^1	twos	2
1 ₂	2^0	ones	1
=	↑	↑	↑
1203 ₁₀	power of two	place	decimal equivalent

MARGIN FIGURE 10-4 Placeholder system, powers of 2 for binary. The example binary number shown is equivalent to the decimal number 1203.

interesting, computers would think the way humans do. It is plausible that a computer could go beyond its specifically programmed instructions to “adapt” to new situations and “learn” from its mistakes. Providing these and other capabilities to computers is the goal of a branch of computer science called artificial intelligence (AI).^{9,10} Computers in operation at the present time, however, are far more simple than the human mind. They basically do exactly what they are programmed to do, no more and no less. Hence the tasks that computers are currently asked to perform must be dissectable into simple, completely defined steps.

Computers are able to perform numerical operations such as addition, subtraction, multiplication, and division through the use of electronic components called “gates.” The key to understanding how gates work is to examine the way that electronic signals are modified in passing through them. At the input end of a gate the two impinging signals are either “high” (e.g., 5 V), representing a binary number 1, or “low” (e.g., 0 V), representing a zero. The function of the gate is to alter the voltage in a fashion that mimics a mathematical or logical operation. The alteration is illustrated by the “truth table” for the gate showing the output values achieved for all possible combinations of input values. For example, Margin Figure 10-8 shows two gates, an AND gate and an exclusive-OR (X OR) gate. The inputs are labeled *a* and *b*. The truth table for the AND gate shows that the output will be 1 (high voltage) if and only if both of the inputs are 1. The truth table for the X OR gate shows that the output will be 1 if and only if the two inputs are different (either *a* = 1, *b* = 0 or *a* = 0, *b* = 1).

Gates and combinations of gates function as mathematical and logical operators for binary numbers. For example, the X OR and the AND gate, when connected as shown in Margin Figure 10-9, produce a truth table that corresponds to the addition operation for binary numbers. When the inputs are added, the output is a 2-bit representation of the result. Adding 1 and 0 in either order produces the binary number 01. When both inputs are 1, the output is the binary number 10 (decimal number 2). Other combinations of gates carry out operations such as subtraction and multiplication, as well as logical operations such as “If *a* is true (i.e., high voltage), then the result (output of the gate) is true.” A large assembly of gates is capable of attaining a high degree of complexity.

It may be difficult to understand how a problem such as the reconstruction of an image from input signals is related to simple gates. Nevertheless, this operation can be traced to individual gates that are actuated in sequence to arrive at an appropriate output in binary form.

Bits, Bytes, and Words

Information in a computer is handled in chunks of various sizes. The fundamental unit of information is the binary digit, contracted to the term “bit,” which refers to the individual ones and zeros that make up a binary number. In this context a one or a zero is considered as a single bit of information. Eight bits make up a “byte” of information, and 2 or more bytes make up a “word” of information. If in a particular type of computer the “word length” is 16 bits, then a word is composed of 2 bytes (Margin Figure 10-10). Common word lengths are 16 bits (a 2-byte word), 32 bits (a 4-byte word), and 64 bits (an 8-byte word). The size of the chunk of information that a particular computer can handle at one time determines the precision of numerical operations and the amount of storage the computer can maintain.

Certain prefixes have become standard for specifying large numbers of bits, bytes, and words. Because the information computers deal with is routinely divided into powers of 2, the prefixes kilo and mega in computer terminology invoke the powers of 2 that are nearest to the quantities 1000 and 1,000,000, respectively. The number 2 raised to the tenth power equals 1024 and is referred to as one “K” or kilo. Thus, 5 Kbytes is actually 5 × 1024 = 5120 bytes rather than 5000 bytes. Similarly, 2 raised to the 20th power equals 1,048,576 and is referred to as one “M” or mega. So, 10 Mwords equals 10,485,760 words, or 485,750 words more than one might expect from the metric meaning of “mega.” This terminology is used consistently in

computers, as there is little cause for confusion in practice. A computer with 2 Mbyte of internal memory has twice as much memory as one with 1 Mbyte of memory.

Storage of Letters and Symbols

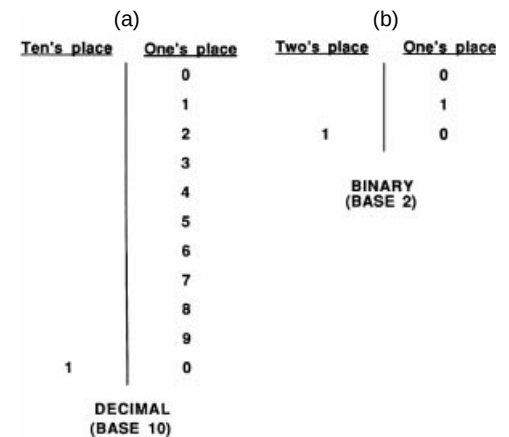
The storage of binary values is accomplished directly by influencing the status (high or low) of voltages in various circuits in the computer. A row of switches that are “high, low, low, high” in voltage might represent the binary number 1001 (decimal 9). Letters and other symbols have no such “natural” representation. Instead, they are recorded in code as specific patterns of binary digits that may be stored and manipulated by the computer. When the computer “communicates” with the outside world (e.g., with a printer), these patterns of bits are converted to the shapes humans recognize as letters and symbols.

The 26 letters of the alphabet, the digits 0 to 9, and a number of special symbols (such as \$, #, &, ...) are referred to collectively as the “alphanumeric character set” of the computer. Depending on the number of symbols and whether both uppercase and lowercase letters are allowed, the alphanumeric character set may consist of as many as 128 elements. Table 10-1 shows, as an example, one of the most widely used schemes of encoding alphanumeric information. This scheme is the American National Standard Code for Information Interchange (ASCII).

Image Storage: Pixels

To store images in a computer, the image must be divided into smaller sections called picture elements, or “pixels” (Margin Figure 10-12). Each pixel is assigned a single numerical value that denotes the color, if a color image is stored, or the shade of gray (referred to as the gray “level”), if a black and white image is stored. The image is like a jigsaw puzzle, except that each piece is a square and has a uniform color or shade

The expression “memory” to describe the storage capacity of the computer was suggested by John von Neumann in 1945. The first all-purpose computers marketed by IBM were referred to as “johnniacs.” This acknowledgment reflected von Neumann’s many contributions to computer designs, including the concept of a program (a code that directs computer operations).

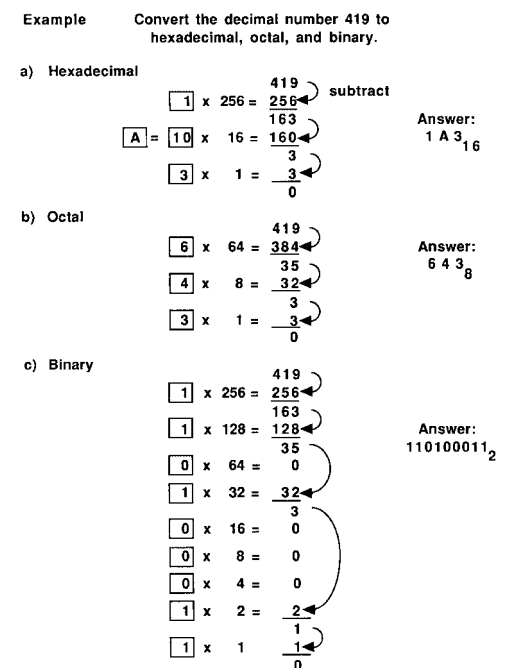


MARGIN FIGURE 10-5

A: Decimal expression of the number 10 involves the use of zero as a placeholder in the one's place and the digit “1” in the ten's place. B: Binary expression of the number 2 involves the use of zero as a placeholder in the one's place and the digit “1” in the two's place.

TABLE 10-1 American National Standard Code for Information Interchange (ASCII)

Character	Binary Code	Character	Binary Code
A	100 0001	0	011 0000
B	100 0010	1	011 0001
C	100 0011	2	011 0010
D	100 0100	3	011 0011
E	100 0101	4	011 0100
F	100 0110	5	011 0101
G	100 0111	6	011 0110
H	100 1000	7	011 0111
I	100 1001	8	011 1000
J	100 1010	9	011 1001
K	100 1011		
L	100 1100		
M	100 1101	Blank	010 0000
N	100 1110	.	010 1110
O	100 1111	(	010 1000
P	101 0000	+	010 1011
Q	101 0001	\$	010 0100
R	101 0010	*	010 1010
S	101 0011	)	010 1001
T	101 0100	-	010 1101
U	101 0101	/	010 1111
V	101 0110	,	010 1100
W	101 0111	=	011 1101
X	101 1000		
Y	101 1001		
Z	101 1010		



MARGIN FIGURE 10-6

Conversion of decimal numbers to hexadecimal, octal, and binary.

Decimal (Base 10)		Binary (Base 2)					Octal (Base 8)		Hexadecimal (Base 16)		
place -	tens	ones	sixteens	eights	fours	twos	ones	eights	ones	sixteens	ones
power of base -	10 ¹	10 ⁰	2 ⁴	2 ³	2 ²	2 ¹	2 ⁰	8 ¹	8 ⁰	16 ¹	16 ⁰
		0					0		0		0
		1					1		1		1
		2				1	0		2		2
		3				1	1		3		3
		4			1	0	0		4		4
		5			1	0	1		5		5
		6			1	1	0		6		6
		7			1	1	1		7		7
		8		1	0	0	0	1	0		8
		9		1	0	0	1	1	1		9
1	0			1	0	1	0	1	2		A
1	1			1	0	1	1	1	3		B
1	2			1	1	0	0		4		C
1	3			1	1	0	1		5		D
1	4			1	1	1	0		6		E
1	5			1	1	1	1		7		F
1	6		1	0	0	0	0	2	0		10
1	7		1	0	0	1	1	2	1		11
1	8		1	0	0	1	0	2	2		12
1	9		1	0	0	1	1	2	3		13
2	0 ₁₀		1	0	1	0	0	2	4		14

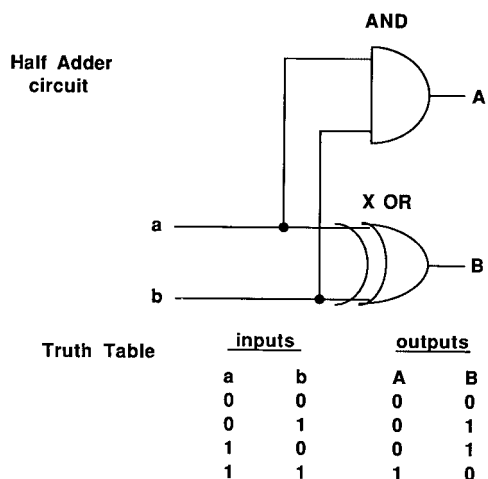
MARGIN FIGURE 10-7

Representation of the decimal numbers 1–20 in other bases.

Operation	Symbol	Truth Table
AND		a b out
		0 0 0
		0 1 0
		1 0 0
		1 1 1
exclusive - OR (X OR)		a b out
		0 0 0
		0 1 1
		1 0 1
		1 1 0

MARGIN FIGURE 10-8

Logic gates convert the input signals a and b to an output signal.

**MARGIN FIGURE 10-9**

The AND gate and the exclusive-OR (XOR) gate, when connected as shown here, produce a 2-bit output that is identical to the binary addition of the a and b signals.

of gray. The numerical values are represented in binary (or octal or hexadecimal) notation, and the image is said to be “digital.”

The set of pixels composing the image is referred to as the image “matrix.” That is, a digital image consists of a matrix of pixels.

Some images are computer generated directly in pixel or digital form. Computed tomography (CT), magnetic resonance, and computed radiography are examples of technologies that produce digital images directly. An image such as a chest film that was not originally “digital” can be entered and stored in a computer by dividing it into a matrix of squares and assigning the average color (or shade of gray) within each square as a single numerical value for the pixel (Margin Figure 10-13). An image modified in this fashion is said to have been “digitized.”

If a few pixels are used in digitizing an image, the result is a coarse or “blocky” appearance. If a larger number of pixels is used, the observer may not even notice that the image is digital (Margin Figure 10-14). It is not necessarily better to break an image into the largest matrix size (smallest pixels). If the computer has limited storage or if the processing and transmission time for the images is restricted, then a smaller matrix size (larger pixels) may be desirable. Matrix sizes are typically powers of 2, reflecting the binary nature of computer storage (e.g., 64, 128, 256, 512, 1024, ...).

Example 10-1

A digital image is constituted as a 128×128 matrix of pixels. Transmission of the image from one computer workstation to another requires 13.7 sec. If the matrix size is changed to 256×256 , how long will it take to transmit the image?

- The 128×128 image consists of $128^2 = 16,384$ pixels.
- The 256×256 image consists of $256^2 = 65,536$ pixels.
- There are $65,536/16,384 = 4$ times as many pixels in the image with the larger matrix size.

Note: Increasing the length of one side of a matrix by a factor of 2 increases the total number of pixels by a factor of 4 since image matrices are usually square.

Therefore the new transmission time is

- $4 \times 13.7 \text{ sec} = 54.8 \text{ sec}$

In this example, the transmission time depends strictly upon the number of pixels. We will see in another section that some data compression techniques can decrease the transmission time.

There is no universal standard code for pixel values as there is for alphanumeric symbols. On the contrary, the ability to reassign numerical values that signify gray levels or colors is one of the most useful principles underlying image manipulation with a computer. A range of numbers can be assigned to any image, for example, with lower numerical values representing darker and higher numerical values representing lighter shades of a gray-scale image. The range of values affects the appearance of the image as shown in Margin Figure 10-15.

The number of bits used to describe the gray level is referred to as the “bit depth” of the image. This terminology comes from thinking of the digital image not as a two-dimensional object but as a three-dimensional block with the gray level as the third dimension (Table 10-2; see also Margin Figure 10-12).

Example 10-2

How many bits are required to store a 512×512 digital image with a bit depth of 8 bits?

- $512 \times 512 \times 8 = 2,097,152$ bits
- Recall that $2^{20} = 1,048,576 \text{ bits} = 1 \text{ Mbit}$
- Therefore $\frac{2,097,152}{1,048,576} = 2$ Mbit are required to store the image

TABLE 10-2 Number of Bits Required to Record a Given Number of Unique Values^a

Number of Bits	Possible Values				Number of Values
1	0	1			2
2	00	01	10	11	4
3	000	001	010	011	8
4	100	101	110	111	16
	0000	0001	0010	0011	
	0100	0101	0110	0111	
	1000	1001	1010	1011	
	1100	1101	1110	1111	

^a The number of unique values that may be recorded equals 2 raised to the number of bits used to record the values.

The problem may be solved without a calculator by noting that

$$512 = 2^9 \quad \text{and} \quad 8 = 2^3$$

Therefore, the total number of bits is

$$2^9 \times 2^9 \times 2^3 = 2^{21} \text{ bits}$$

(superscripts add when multiplying powers of the same base number), but

$$\frac{2^{21} \text{ bits}}{2^{20} \text{ bits/Mbit}} = 2^1 = 2 \text{ Mbits}$$

(superscripts subtract when dividing powers of the same base number).

Example 10-3

How many bytes are required to store the above image?

$$\begin{aligned} \frac{512 \times 512 \times 8 \text{ bits}}{8 \text{ bits/byte}} &= 512 \times 512 \\ &= 2^9 \times 2^9 \\ &= 2^{18} \text{ bytes} \end{aligned}$$

In this example, each pixel is 1 byte “deep,” so the number of pixels is the same as the number of bytes.

Example 10-4

How many bytes are required to store the above image if the bit depth is reduced to 6? Increased to 9?

A byte takes up 8 bits. By using only 6 bits we are simply not filling each byte. But unless some special memory configuration is used, the basic unit of information is the byte. So, each pixel cannot take up less than 1 byte. Therefore, even though the bit depth may be less than 8, 1 byte per pixel is still required. So for a bit depth of 6, the number of bytes required is

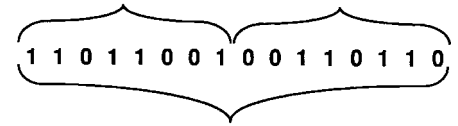
$$512 \times 512 \times 1 \text{ byte} = 2^{18} \text{ bytes}$$

If the bit depth is any number between 8 and 16, 2 bytes per pixel are required. Therefore, if the bit depth is increased to 9, the number of bytes required is

$$512 \times 512 \times 2 \text{ bytes} = 2^{19} \text{ bytes}$$

8 BITS = BYTE

BYTE



WORD

MARGIN FIGURE 10-10

A byte is composed of 8 binary digits, or bits. A word is composed of an integral number of bytes. The word shown here consists of two bytes, or 16 bits. However, word length is not universally recognized, so there are 16-, 32-, 64-, and so on, bit words.

Power of 10
Nearest Power of 2 (prefixes or “units”)

1,000 = 10³

1,024 = 2¹⁰ = 1 K (“kilo” or “kilo”)

1,000,000 = 10⁶

1,048,576 = 2²⁰ = 1 M (“Mega”)

1,000,000,000 = 10⁹

1,073,741,824 = 2³⁰ = 1 G (“giga”)

1,000,000,000,000 = 10¹²

1,099,511,627,776 = 2⁴⁰ = 1 T (“tera”)

1,000,000,000,000,000 = 10¹⁵

1,125,899,906,842,624 = 2⁵⁰ = 1 P (“peta”)

1,000,000,000,000,000,000 = 10¹⁸

1,152,921,504,606,846,976 = 2⁶⁰ = 1 E (“exa”)

Really big numbers:

1, ... (80 zeros) ... = 10⁸⁰

~ 1.186 × 10⁸⁰ = 2²⁶⁶ = 17

1, ... (100 zeros) ... = 10¹⁰⁰

~ 1.750 × 10¹⁰⁰ = 2³³³ = 1 gogol

Examples of files that would occupy 1-1000 of these units (in bytes)

text files

a medical image
or a book

short term, on-site storage
of medical images

One year of image storage
in an “all digital” hospital

10 year medical image storage
for an “all digital” hospital

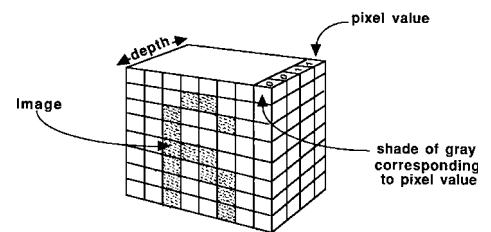
All of the angiographic studies
ever performed

Estimated number of protons in
the Universe

The biggest number formally named.
(except for the gogolplex, which is 10⁸⁰⁸⁰)
(and except for 2 gogolplex, 3 gogolplex, etc.)

MARGIN FIGURE 10-11

Powers of 10 and prefixes.



MARGIN FIGURE 10-12

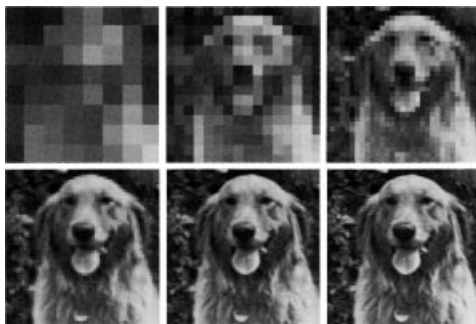
A digital image may be thought of as a three-dimensional object composed of a number of small cubes, each of which contains a binary digit (bit). The image appears on the front surface of the block. The “depth” of the block is the number of bits required to describe the gray level for each pixel. The total numbers of cubes that make up the large block is the number of bits required to store the image.

Two mathematicians, Alan Turing and Kurt Gödel, were among those who made substantial contributions to the logic underlying computer design. Both were included in *Time* magazine's (March 29, 1999) list of Greatest Scientists and Thinkers in the twentieth century. Turing set standards for artificial intelligence and Gödel set limits on what could be computed within a given logical framework.



MARGIN FIGURE 10-13

The photograph on the left has been digitized for the purpose of manipulation and storage in a computer.



MARGIN FIGURE 10-14

As the matrix size of a digital image is increased, the relative size of each pixel decreases and the image appears less "blocky." The images shown here, beginning at top left, consist of 8×8 , 16×16 , 64×64 , 128×128 , 256×256 pixels. Each image has a bit depth of 8, corresponding to 256 shades of gray.



MARGIN FIGURE 10-15

As the bit depth of a digital image is increased, the number of shades of gray that are stored for each pixel is increased. The images shown here are, top left, 1 bit (2 shades), 2 bit (4 shades), 3 bit (8 shades), 4 bit (16 shades), 6 bit (64 shades), 8 bit (256 shades). Each image has a matrix size of 768×512 pixels.

Analog-to-Digital and Digital-to-Analog Converters

Measurable quantities may be separated into two categories, analog and digital. Analog quantities are continuously variable and can take on any value between two limits. The list of possible values is limited only by the tools of measurement. Digital quantities are discrete and can assume only specific values separated by gaps. For example, the height of an individual is an analog quantity, but the number of people in a classroom is a digital quantity. Time can be measured in analog fashion with an analog watch (a watch with hands) or in a digital fashion with a digital watch (a watch with a display showing several decimal digits).

Computers operate with digital quantities, but much of the information we routinely use is in analog form. Therefore, conversion from one form of information to the other often is necessary. For example, electrical signals from some radiation detectors (such as photomultiplier tubes in a gamma camera) are analog. If a computer is to use these signals, an analog-to-digital conversion (ADC) must be performed. A computer used to control devices in a laboratory must produce signals that at least appear to the equipment to be analog. The production of a digital image is a representation of digital information as an analog picture. That is, the presentation of an image to an observer is a form of digital-to-analog conversion (DAC).

When an analog quantity is translated into a digital value, the issue of the precision of the translation arises. For example, suppose that a radiation detector produces a signal of 3.675 V out of a possible range of 0 to 10 V. A computer that uses 3 bits to record the signal can register, at most, 8 unique values (they are 000, 001, 010, 011, 100, 101, 110, and 111). For a range of 0 to 10 V, these eight values differ by increments of $10/8$, or 1.25 (Margin Figure 10-10). An analog quantity between 0 and 1.25 might be recorded as 0, a quantity between 1.25 and 2.50 might be recorded as 1.25, and so on. The value of 3.675 is between 2.50 and 3.75 and would be recorded as 2.5, with an absolute error of 1.175. In this example, the maximum digitization error is the size of the increment (i.e., 1.25 V) because a value of 1.249999 V would be recorded as 0. The maximum error is therefore 1.25 V. If the analog values were randomly distributed, we would expect to see all errors from zero to slightly less than 1.25. Thus, the average digitization error is half the digitization increment, or 0.625 V. We can generalize to the following formulas:

$$\text{err}_{\max} = \frac{R_a}{N} \quad (10-1)$$

where $\text{err}_{\max}$ is the maximum digitization error, R_a is the analog range, and N is the number of distinct values available to the digitizer. Also,

$$N = 2^n, \quad (10-2)$$

where n is the number of bits. Combining Eqs. (10-1) and (10-2) yields

$$\text{err}_{\max} = R_a 2^{-n} \quad (10-3)$$

Then

$$\text{err}_{\text{avg}} = \frac{\text{err}_{\max}}{2} = \frac{R_a}{2^{n+1}} \quad (10-4)$$

where err_{avg} is the average error.

COMPUTER SYSTEM HARDWARE

A computer system needs both hardware and software to function. Hardware is any physical component of the computer; it has mass and takes up space. Software is any set of instructions required to operate the computer or perform mathematical operations.

There are many types of computers. The purpose of this section is to introduce the features that are common to most computers and to illustrate some operating

principles of computers used in medical imaging. A diagram of a computer is shown in Margin Figure 10-17.

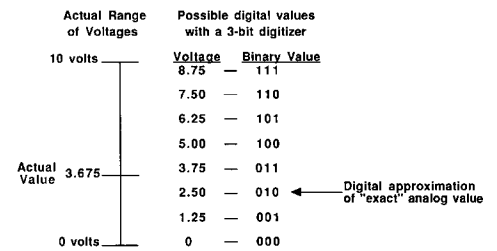
Input and Output

A device that allows information to be entered into the computer is an input device, and a device that transfers information from the computer to the outside world is an output device. Many devices can perform both functions. For example, a disk drive may be used to read information from disks as an input function. The same disk drive may also be used to write information from the computer to a disk for storage, an output function.

A list of devices and their usual roles as input or output devices is shown in Table 10-3. Many of the devices that perform input or output functions also perform storage, interfacing, and networking functions that are discussed in a later section. All of the devices in Table 10-3 may be used in diagnostic imaging. The output devices most used are the display screen and the hard-copy device. Commonly used input devices in diagnostic radiology include digital radiographic equipment, CT gantries, magnetic resonance imaging receiver coils, ultrasound transducers, and gamma cameras. The computer can be considered as the actual imaging device, with the equipment mentioned above serving to gather and transmit data about the patient to the computer.

Central Processing Unit

The central processing unit (CPU) acts as the “traffic cop” of the computer. Its function is so central to the operation that if any one component can be said to be a computer in and of itself, it is the CPU. It determines when different components of the computer are needed to perform required tasks. For example, a simple task such as “add two numbers” requires the following steps: (1) Locate the first number from an input



MARGIN FIGURE 10-16

A 3-bit digitizer is used to record voltage signals over the range of 0 to 10 V. Because only eight digital values are possible for a 3-bit digitizer, one scheme, illustrated here, would assign all voltages between 0 and 1.25 V to the value 0 V (binary 000), all voltages between 1.25 and 2.50 V to the value 1.25 V (binary 001), and so on.

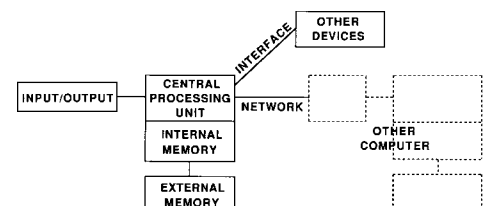
Quantum Computers: In the quest for greater computing power, one area of research involves using some of the properties of objects, such as small numbers of photons, electrons, or nuclei, whose behavior is dominated by quantum mechanical effects.

One such phenomenon is “quantum teleportation” in which a state that could represent one bit of information (such as the direction of electromagnetic fields in a light beam) may be changed instantaneously over large distances. The implications for high-speed communications are obvious.

Another potentially useful property of a quantum mechanical system is that it exists in all possible combinations of states up to the moment of measurement. This property suggests that the various states could represent ones and zeros in computations and that, until the moment of measurement, the “quantum computer” could explore all possible combinations of solutions to a problem. One way to construct such a system of quantum mechanical spins involves the same nuclear magnetic resonance phenomenon in liquid samples that is used in magnetic resonance imaging of the human body.

TABLE 10-3 Sampling of Input/Output Devices

Device	Purpose	Input	Output
Keyboard	Manual entry of alphanumeric data	X	
Light pen	Entry of data by a hand-held wand	X	
Joystick	Manual entry of directional information	X	
Mouse	Manual entry of directional information	X	
Modem	Communication through phone lines	X	X
Hard copy	Conversion of a video signal to a permanent image		X
Display screen	Visual display of information		X
Printer	Provides paper copy of alphanumeric information, graphs, and charts		X
X/Y plotter	Provides continuous lines, graphs		X
Touchscreen	User-interactive display	X	X
Disk/tape drive	External storage	X	X
Digitizer	Used to enter graphs or images directly	X	
Network interface	Communications	X	X
Equipment interface	Communications	X	X

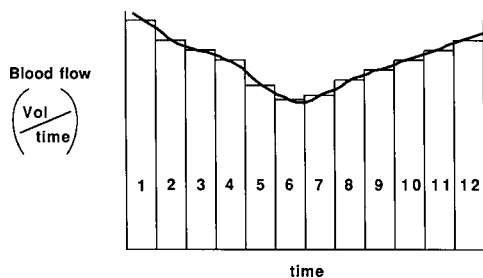


MARGIN FIGURE 10-17

Computer system hardware.

In February 1996, the chess-playing computer Deep Blue defeated World Champion Garry Kasparov. The significance of this victory has been debated ever since in the artificial intelligence community.

Molecular Computers: One approach to the “next generation” of computers is molecular computing. The nucleotide sequence of each DNA molecule could be used to encode a potential solution to a problem, such as the pixel values of a computed tomography image that might correspond to the x rays that were detected by the system. Various techniques in molecular biology could, in principle, be used to recombine the molecules in all possible combinations and select specific sequences that correspond to the best solution to the overall problem. A liter of a solution of DNA molecules contains as many as 10^{20} molecules. If used as active components of a “molecular computer,” it would represent a vast improvement over today’s chips, which typically contain “only” billions of components.



MARGIN FIGURE 10-18

Total area under a blood flow curve represents the total volume of blood. The area may be approximated. The computer estimates the blood volume by finding the area of each of the 12 rectangles and summing the results.

device; (2) store the number in a temporary storage location, and keep track of where it is; (3) locate the second number from an input device; (4) perform the addition function (itself a multistep process that may be “hard-wired” as a circuit such as the adder shown in Margin Figure 10-9 and route the result of the calculation to a third temporary storage location; (5) identify the temporary storage location where the sum is placed; and (6) report the result back to the main program, and look for the next instruction.

A CPU operates on data by assigning memory locations or “addresses” as indicated in the example above. Each address can be a binary (octal or hexadecimal) number no longer than the maximum word length of the computer. As shown earlier, the number of bits determines the number of unique values that a computer may recognize. Thus, the number of unique items that can be referenced by the computer is determined by the number of addresses it can hold at one time. This number is determined by the word length recognized by the computer. The word length is referred to as the “bit depth” of the CPU. If a certain processor has bit depth of 32, for example, it can process 2 to the 32nd power, or 4,294,967,296 instructions.

A 64-bit processor is inherently more powerful than a 32-bit CPU. A processor with greater bit depth is able to handle more storage locations and its numerical operations have a higher precision compared with those performed by a CPU with a smaller bit depth.

Most computers currently in operation use serial processing. This approach requires that one task be completed before another is begun. In many problems, the result of one task is the input for the next, and serial processing is necessary. In some problems, however, the completion of one task is independent of at least some other tasks. In these cases it may be possible for one processor to work on one task while a separate processor works on another. These tasks are said to be performed “in parallel.” A computer that can perform tasks in parallel is termed a “parallel processor.”

Serial and parallel processing are illustrated by the problem of determining the area under the curve in Margin Figure 10-18. This curve could represent the relative volume of blood in the heart at different times as measured by a gated blood pool nuclear medicine study. The area under the curve represents the volume of blood flowing through the heart during a cardiac cycle. In the serial approach to the problem, the computer might “draw” a series of rectangles under the curve and keep a running sum of the area of the rectangles. At the end of the curve, the summed area of the rectangles approximates the total area under the curve. In the parallel approach the computer recognizes that it has the data for the entire curve in memory. Because the areas of the rectangles are independent, the computer could “draw” and compute the area under each of the rectangles simultaneously, provided that it had enough independent processors. The problem could then be solved more rapidly.

At the present time, medical imaging systems do not make extensive use of parallel processors because they are costly and complex. Problems such as the blood flow example above can be solved in less than a second in serial fashion. Unless a large number of such problems must be solved quickly, there is little reason to use parallel processors. These processors are used for problems in other fields that cannot be solved by serial processing in a reasonable time. Some examples include large-scale problems in aerodynamics, meteorology, cosmology, and nuclear physics.¹¹ The use of parallel processors in radiology could increase in the future as imaging applications become more complex.

Special Processors

The array processor is a special-purpose processor that can be considered a small-scale parallel processor. An array processor uses a single instruction to perform the same computation on all elements of an array of data. Tremendous savings in computer time are achieved by using array processors to manipulate digital images. Such processors are routinely used in CT and magnetic resonance imaging units and are becoming standard in all types of digital imaging systems.

Another special processor that is used routinely in radiology is the “arithmetic processor.” Many mathematical functions such as exponential values, square roots, and trigonometric functions can be calculated in clever but time-consuming fashion by repeated application of procedures such as additions that are “hard-wired” in gate circuits. The CPU of a typical computer is designed to perform alphanumeric and logical operations as well as handle mathematical operations. A faster (and usually more accurate) alternative involves the use of an arithmetic processor, a device that executes fewer steps in performing certain mathematical equations. The arithmetic processor is optimized to perform mathematical functions only. It is called upon by the CPU to provide these functions and return the results to memory.

Speed: MHz, MIPS, and MFLOPS

The speed of a CPU may be specified in a number of ways. One of the most fundamental specifications is the “clock rate.” As the CPU executes the various steps required to complete a task, it keeps synchrony among the various components (input-output devices, memory, etc.) with a “master clock.” The minimum time required to execute an instruction is the clock’s “cycle time.” The inverse of this cycle time, the theoretical upper limit of the number of instructions the CPU can execute per second, is the clock rate. Because the clock rate is a frequency in cycles per second, it is commonly expressed in multiples of hertz, such as megahertz (MHz) or gigahertz (GHz). For example, a CPU having a cycle time of 1.25 nanoseconds has a clock rate of

$$\begin{aligned}\frac{1 \text{ instruction}}{1.25 \times 10^{-9} \text{ seconds}} &= 800 \times 10^6 \text{ instructions per second} \\ &= 800 \times 10^6 \text{ hertz} \\ &= 800 \text{ MHz}\end{aligned}$$

Currently, CPUs in personal computers have a maximum clock rate of 1.5 gigahertz. Because of economies of scale in sale and manufacture, CPUs in computers associated with medical devices are the same CPUs that are sold commercially in quantity to PC users.

The CPU may take several clock cycles to execute an instruction. Therefore, a more realistic measure of speed of a CPU is the number of instructions per second that it can execute. For modern computers, this is often specified in millions of instructions per second (MIPS). Today’s PCs can perform up to 100 MIPS. Some special purpose machines that have been built to test the limits of design can perform up to 10^{12} (prefix “tera”) instructions per second (TIPS).

In numerical calculations it is often convenient to specify numbers in scientific notation (e.g., 1.25×10^{-9}), with a mantissa (e.g., 1.25) specifying a numerical value and an exponent (e.g., -9) specifying the location of the decimal point. This specification facilitates operations on numbers that cover a wide range of values. In computer jargon, scientific notation is referred to as “floating point.” Several bits of a floating-point word are used for the mantissa and several others for the exponent. The exponent is stored as a power of two, but the principle is the same. A few more bits are required for the signs of the mantissa and exponents. The instructions for floating-point arithmetic are more complicated (and therefore time-consuming) than for integer arithmetic, but the use of floating point is often necessary for scientific calculations. One million floating-point operations per second is abbreviated as 1 MFLOPS.

Supercomputers with parallel processing capability can attain processor speeds of 1000 MIPS and floating-point speeds of over 300 MFLOPS. So-called “massively parallel” systems with several hundred microprocessors achieve giga-FLOP speeds.¹²

The speed with which the entire computer can complete a task is usually limited by things other than the CPU. The rate at which the device can transfer data, access disk drives, or transfer information to peripheral devices may play a major role in determining how quickly it can accomplish complex tasks such as digital image

processing and display. For this reason, overall computing speed is often measured by the time required to run standard programs successfully. These standard programs, called “benchmark” programs, are selected on the basis of their representation of the particular tasks that the computer in question would be expected to perform.

Memory

Memory in a computer can be divided into internal memory and external memory (or external storage). Internal memory is actively used by the computer during execution of a program, and it is directly accessible during an instruction cycle. External memory requires that the computer read information from an external storage device such as magnetic tape or disk. An analogy can be made with the student taking notes in a course. If the student knows the material without looking at the notes, the formation is stored in “internal memory.” If the student must refer to the notes, then the information is in “external memory,” or storage.

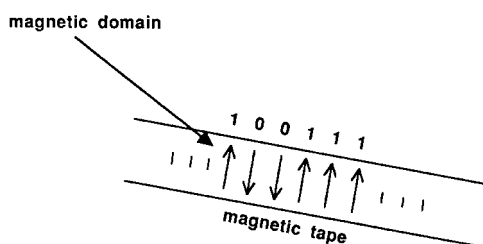
There are many types of internal memory. Random-access memory (RAM) is used as a scratch pad during calculations. Data can be stored there, read out, and replaced with new data. RAM is available when the computer is on. It contains information placed there for the problem that the computer is working on at the moment. Most RAM is “volatile,” meaning that the information is lost when the computer is turned off. Read-only memory (ROM) is not volatile. It is used to store sets of instructions that the computer needs on occasion. The information in ROM is “hard-wired” (i.e., it is a physical part of the electronic components) and cannot be changed by the computer (hence the term “read only”). A variation of ROM, called programmable read-only memory (PROM), can be changed by exposure to an external electrical or optical device such as a laser so that reprogramming of the PROM is possible but not routine.

In a digital imaging device the internal memory must be large enough to contain at least one digital image. A $512 \times 512 \times 8$ -bit image requires $\frac{1}{4}$ Mbyte. Therefore, an imaging device that manipulates several such images needs several Mbytes of internal memory.

External memory devices are used to store images that are not needed for an immediate imaging task. There are various levels of separation from the computer. Images for which the diagnosis has been made are often stored for several years for legal purposes. These images are said to be “archived” (e.g., on magnetic tape), and they have to be remounted on the computer to retrieve the information. Images that are on hard disk drives are physically mounted on the computer and can be accessed within milliseconds. In magnetic storage devices the individual bits are recorded as magnetic “domains” where “north pole up” might mean a 1 and “north pole down” might mean a 0 (Margin Figure 10-19).

The most commonly used external memory devices store information in one of two ways. Magnetic tape, a mainstay of computer storage into the 1970s in the form of large reel-to-reel devices, now plays a role as a relatively cheap “off-line” storage option in the form of smaller closed cassettes. Although it may take a few seconds or even minutes to load the modern tape cassette and forward it to the location where the data of interest are stored, the long-term cost of buying and maintaining this storage medium are low.

Various optical storage devices are also available. The time required to access a particular string of data on an optical storage device may be nearly as short as the time required to access data on a (magnetic) hard disk. Optical disks contain tiny imperfections in an otherwise smooth mirror-like surface. A hole burned into the surface of the disk may be used to indicate a binary 1, while the absence of a hole indicates a 0. Optical disks that are created by burning are examples of “write once, read many” times (WORM) devices. This feature makes them suitable for long-term archival storage while still retaining a greater access speed. Disks 12 cm in diameter have become standard for musical recording and for routine PC use. These disks are known as “compact disks” or CD ROMs. Higher-capacity disks of the same size as



MARGIN FIGURE 10-19

Magnetic storage media such as magnetic tapes utilize magnetic “domains,” regions of magnetization that are allowed two possible configurations to store binary digits.

TABLE 10-4 Characteristics of Storage Media

Media	Capacity	Access Time
Magnetic tape	Gbyte	sec-min
Floppy disks	2 Mbyte	msec
Hard disks	10 Gbyte	msec
Optical disks	5 Gbyte	msec

CD ROMs, but capable of storing enough digital information to hold several hours of high-quality video, are known as digital video disks (DVDs). DVDs are becoming more prevalent. Mechanical devices are sometimes used to fetch and mount disks on demand from a large collection of optical disks. These devices are known as “jukeboxes.” Table 10-4 shows some of the characteristics of commonly used external storage media.

■ SOFTWARE

The term *software* refers to any set of instructions the computer uses to function effectively. Specific sets of instructions are necessary for any computer. The “operating system” consists of instructions that govern the use of input and output devices, set aside internal storage locations, direct the computer in tasks such as copying, naming, and erasing files from external storage, and perform other necessary tasks. The operating system always must be executed before any other programs are run. In many computers the operating system program is executed automatically when the power is turned on. Some of the operating system instructions are usually resident in ROM or PROM. Running the operating system programs is referred to as “booting” the computer.

Software that has been written to perform a specific task, such as to calculate ejection fractions for a nuclear cardiology study, is called a “program.”

Several levels of software are usually present when a computer is operating, and instructions may be written in various “languages.” High-level languages are most accessible to the typical programmer. They are written in statements that use everyday language such as “go to, return, and continue.” In a high-level language, mathematical formulas are expressed just as they would be written on paper. Assembly language is a step closer to direct communication with the computer. Statements in assembly language direct the computer to perform specific tasks such as to store data in certain locations. High-level languages such as Visual BASIC, C++, or SQL tell the computer what to do in a general way. Assembly languages give the computer specific instructions.

Both high-level and assembly languages must ultimately be translated into machine language. Machine language is represented simply as strings of ones and zeros. It directs the fundamental setting and resetting of various electronic components that yield signals sent to an output device. Assembly language programs are converted into machine language by a translation program called an “assembler.” Higher-level languages may be converted into machine language in one of two ways. They may be translated all at once into machine language by a program called a “compiler,” or they may be translated line by line each time the program is executed by a program called an “interpreter.” The interpreter is used when the programmer needs to make changes in the program. Once the high-level language is compiled, the program cannot be changed, because individual lines in the program are no longer accessible. The compiler is used when further changes are not required, and speed of execution is important. Compared with interpreted programs, compiled programs usually require much less time to execute. Different types of software and their interrelationships are compared in Margin Figure 10-20.

■ NETWORKING

The increased use of computers in all areas of medicine, including diagnostic imaging, creates the desire to link the computers together, thereby making information present in one computer available to all computers in an institution or community. This linking of computers, called “networking,” is a topic of current interest.^{13–16}

In the late 1960s, the U.S. Department of Defense wanted to develop a computer network that would survive a nuclear attack. The network would continue to deliver

The term “booting”, meaning the startup process for a computer, comes from the phrase “to pull oneself up by the bootstraps” (i.e., to be a self-starter).



Jukebox: Juke is a term from Gullah, a language spoken by descendants of West African slaves living on islands off the coast of Georgia and South Carolina. Juke, one of the few terms from Gullah to be adopted into common English usage, means bad, wicked, or disorderly. It was used to refer to roadside drinking establishments in the Southeastern United States beginning in the 1920s that came to be known as “juke joints.” When mechanical devices capable of selecting and playing individual selections of music from a collection of vinyl disks became available, the devices, often located in “juke joints,” became known as “jukeboxes.” Today, this term is used to refer to mechanical devices for selecting and interfacing multiple digital storage disks (CD-ROM, Optical, Magneto-Optic, etc.). The earlier connotation of wickedness is not generally associated with the information management specialists who use these devices.